

Rethinking the Arithmetic of Multi-Agent Medical Consultation

Anonymous ACL submission

Abstract

In medicine, agreement is evidence. When independent specialists converge on a diagnosis, the convergence itself is information, and that is why tumour boards and second opinions change decisions. Multi-agent consultation has become the default design pattern in medical language-model systems on the strength of that inheritance. We test whether it survives the transfer, across 420 multi-agent configurations and 144,499 episodes over seven models, six independently trained ecosystems, three benchmarks and five architectures. Agreement survives; the independence that gave it meaning does not. A nine-member panel speaks with one voice on 69.5% of items and is wrong on 39.7% of the items it agrees on. Peer discussion raises unanimity to **98.3%** and lowers the accuracy of that unanimity to 54.3%: the panel converges on an answer, not on a correct one. Agreement separates the panel’s right answers from its wrong ones with $AUC = 0.59$, and self-reported confidence with Youden $J \leq 0.18$. The cause is measurable. The mean pairwise error correlation is $\varphi = 0.79$, leaving nine members worth 1.22 **effective independent opinions**; discussion drives it to 0.990, so a consultation at the moment of consensus is worth 1.01 physicians. Independence cannot be bought: normalising φ by its attainable maximum shows the apparent decorrelation from mixing capability is an artefact of unequal error rates, and every pair we can form across six ecosystems—three American and three Chinese—moves a nine-member panel from 2.60 effective opinions to 2.29, and at matched capability from 2.31 to 2.02: the wrong way. Nor can any rule read past it. On a scale where picking a member at random scores 0 and always picking a correct one scores 100, the five architectures score -12 to $+4$. A medical panel that agrees is not a panel that knows. Its consensus must not reach a clinician as confidence.

1 Introduction

Ask nine AI specialists a hard clinical question and let them confer. They will reach unanimity 98.3% of the time and be wrong on 45.7% of the items they agree on. By the moment they agree they carry 1.01 independent opinions between them.

In medicine that combination is not a disappointment but a hazard. Concordance among independent specialists is evidence—it is why a tumour board’s agreement moves a treatment decision, and why a second opinion that matches the first ends the question. A panel that agrees almost always, and is wrong on nearly half of what it agrees on, supplies the form of that evidence and none of its content.

Now ask a different question of the same panel: was *anyone* in it right? Averaged over the 21 model–benchmark cells at $n_a=9$, some member holds the correct answer 60.4% of the time against 50.7% for one doctor working alone. The panel returns 52.2%.

A consultation is worth convening only if two things are true. Somebody in the room must be right when the others are wrong, and the room must be able to tell who. This paper measures both. Panels of language models have a little of the first and almost none of the second, and every failure reported here follows from that pair.

Multi-agent “consultation” has become the default design pattern in medical language-model systems. MedAgents (Tang et al., 2024), MDAgents (Kim et al., 2024), MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) all instantiate the same clinical intuition. A panel of specialists should diagnose better than one generalist, the reasoning goes, because a tumour board diagnoses better than one physician. The pattern is being deployed faster than it is being measured.

The intuition is not merely an analogy; it is an inheritance. Pooling diagnostic judgements

improves accuracy in human medicine for a specific reason, namely that physicians trained differently make different mistakes (Kurvers et al., 2016; Kämmer et al., 2017). Majority voting converts that independence into accuracy. Strip the independence away and the arithmetic stops working, however many voters remain. Nobody has checked whether language-model panels satisfy the premise they borrow.

They do not. Measuring the pairwise error correlation between panel members across 420 multi-agent configurations, we find $\varphi = 0.79$: agents prompted into different medical specialties fail on the same items. A nine-member panel therefore carries 1.22 effective independent opinions, and peer discussion drives the correlation to 0.990, at which point a nine-agent consultation is informationally equivalent to 1.01 physicians. The panel does not converge on an answer. It collapses onto one.

That is the first pillar, and it is the one prior work would predict. It also survives the obvious remedy. Reaching across separately trained families inside one vendor lifts a nine-member panel from 1.22 to 2.60 effective opinions; reaching across the vendor boundary—to Google, Anthropic, DeepSeek, Alibaba and Zhipu—gives 2.29. Two models built by different companies on different corpora fail on the same medical items about as reliably as two checkpoints from one lab.

The second pillar is less expected. Independence is scarce but not absent: the 9.3 pp between one doctor and the panel’s oracle is real signal, held by some member on some items. On a scale where picking a panel member at random scores 0 and always picking a correct one scores 100, majority voting, an auditing orchestrator, peer debate and tiered referral score between 1 and 9. No architecture we test converts what the panel knows into what the panel says, because the quantity such a conversion needs—a signal that separates a right member from a wrong one—is missing: self-reported confidence separates them with Youden $J = 0.166$.

Two observable consequences follow, and they are the same fact seen twice. Consensus carries almost no information, because it is agreement among copies: unanimity separates the panel’s right answers from its wrong ones at $AUC = 0.59$, and deliberation moves unanimity up and its accuracy down. And panel size predicts nothing, because tripling the panel adds a tenth of an opinion.

Establishing this requires a grid large enough to separate architecture from panel size from task difficulty: 420 multi-agent configurations and 144,499 episodes across seven models from six independently trained ecosystems, three medical benchmarks, five canonical architectures, panel sizes $N \in \{1, 3, 5, 7, 9\}$, and a budget-matched single-model control at every cell. Medicine makes the measurement possible in a way general-domain agentic benchmarks cannot: multiple-choice clinical reasoning uses no tools, so coordination can be measured without the confound of tool-call overhead. Scaling principles for general-domain agent systems (Kim et al., 2026) supply the architecture taxonomy and the controlled-comparison design our grid is built on; the error-correlation and consensus-calibration measurements that carry our argument are new here.

2 Related work

Why pooled judgements help, and what they require. Collective-intelligence results in medicine rest on error independence: pooling independent diagnostic judgements improves accuracy (Kurvers et al., 2016; Wolf et al., 2015), plateauing at roughly five raters (Kämmer et al., 2017), and groups of nine outperform individuals on the Human Dx platform (Barnett et al., 2019), the clinical instance of a general principle (Surowiecki, 2004) that multidisciplinary tumour boards institutionalise (Radcliffe et al., 2019). The framing question is not whether two heads beat one but when (Hautz and Kämmer, 2024). The same literature documents what happens when independence fails: conformity to a unanimous majority (Asch, 1956) and groupthink in deliberating groups (Janis, 1972). Correlated errors across models bound ensemble benefit in general (Kim et al., 2025), and multi-agent debate can converge on confident-but-wrong answers (Zhu et al., 2026). This paper measures that failure directly in clinical panels, where diagnostic error is already common enough in ordinary practice (Graber, 2013; Singh et al., 2014) that manufactured agreement is a safety concern rather than an efficiency one. The strongest evidence that the independence premise is what matters comes from the hybrid case: pooling 40,762 physician differentials with five frontier models over 2,133 open-ended vignettes, Zöller et al. (2025) find that human–model collectives beat physician collec-

tives *and* model ensembles, and attribute the margin to humans and models making different kinds of errors. Their result is the consequence; the quantity that produces it is the one we measure. It also establishes the regime: independence arrives when the added judgement comes from outside the model ecosystem, which is the boundary our diversity ladder (§4.5) locates from the inside.

Medical language models and multi-agent systems. Medical question answering became a capability benchmark once models matched licensing-exam thresholds (Kung et al., 2023; Nori et al., 2023; Singhal et al., 2023, 2025), and later work moved to conducting encounters (Tu et al., 2025; McDuff et al., 2025). Prompting strategy carries much of that progress (Nori et al., 2024), which is why architectural claims need budget-matched controls. On the multi-agent side, MedAgents (Tang et al., 2024) introduced role-played specialist collaboration, MDAgents (Kim et al., 2024) added adaptive routing, and MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) formalised teamwork components, the latter sweeping team sizes 2–5. Every published medical ablation stops at $N \leq 5$, none reports a dose-response curve, and none runs a budget-matched single-model control. Skeptical evaluations find that orchestration often fails to beat well-prompted single models at substantially higher cost (Zhu et al., 2025; Liu et al., 2026; Zhao et al., 2025); our independence measurement supplies the mechanism those results lack. The pattern has since diversified along every axis except the one measured here. Chen et al. (2024b) and Zhou et al. (2025) specialise the roster—an MDT for rare disease, role-assigned radiologist and generalist agents for multimodal input—and Liu et al. (2025) adds a director to coordinate them. Ma et al. (2025) replaces majority voting with logical contradiction resolution and Li et al. (2025) lets agents accumulate diagnostic knowledge across a cohort, both of which are attempts to escape a voting rule rather than to escape the correlation that makes voting weak. Fan et al. (2024), Sanghvi et al. (2026) and Sviridov et al. (2025) move the setting itself, from single-shot selection to simulated, interactive and multimodal encounters where agents must ask before they answer, and Chen et al. (2026) and Shi et al. (2026) carry the pattern into inpatient pathways and outcome prediction, where triage, diagnosis and treatment are separate coordinated

stages. These are complementary: each varies the aggregation rule, the roster or the interaction protocol, and each is a candidate for raising the capture rate we define in §4.6. None of them measures the error correlation among its own agents, so none can say whether the signal its mechanism is competing for is present.

Closest to our central question, Yuan et al. (2026) compare single-vendor and mixed-vendor panels of three frontier models on rare-disease diagnosis and report that mixed-vendor teams win, attributing the gain to complementary inductive biases surfaced by an overlap analysis. That is the same lever we isolate in Table 6, measured with a different instrument and in the regime our results predict is the favourable one—three models of comparable strength, where decorrelation carries no quality penalty. Two of our findings bear on how such comparisons should be read. Raw overlap and raw correlation are depressed mechanically when members differ in accuracy (Eq. 3), so vendor-diversity effects should be reported normalised; and the lever is small wherever we can measure it, moving a nine-member panel from 1.22 to 2.60 effective opinions within one vendor and to 2.29—not further, and in fact back—once the panel spans six ecosystems on two continents.

Scaling agents, and scaling inference. Debate protocols improve factuality (Du et al., 2024; Liang et al., 2024) and frameworks such as MetaGPT (Hong et al., 2024) and AutoGen (Wu et al., 2024) made the pattern easy to deploy (Zhang et al., 2024). On how far it scales, “more agents is all you need” (Li et al., 2024) reported monotone gains, while Chen et al. (2024a) showed the relationship is non-monotone and Li et al. (2026) confirmed an inverted-U in agent count; Yang et al. (2026) attributed the shape to opinion diversity, showing two diverse agents can match sixteen homogeneous ones. Kim et al. (2026) fit a predictive scaling principle across architectures, model families and agentic benchmarks, and supply the architecture taxonomy and controlled-comparison design our grid is built on. Test-time compute offers the alternative to adding agents (Kaplan et al., 2020; Hoffmann et al., 2022; Brown et al., 2024; Snell et al., 2024), and self-consistency (Wang et al., 2023) over chain-of-thought (Wei et al., 2022) is the budget-matched control we run at every cell.

Confidence and calibration. Our safety analysis depends on whether stated confidence is informative. Models are partially aware of what they know (Kadavath et al., 2022), but elicited confidence is weakly calibrated and sensitive to how it is asked (Xiong et al., 2024; Guo et al., 2017), and clinical evaluations report fluent but unreliable responses (Goodman et al., 2023) with demographic biases that propagate (Omiye et al., 2023; Zack et al., 2024). We add a coordination-specific finding: deliberation degrades the discriminative power of confidence rather than improving it.

3 Agent systems and tasks

3.1 System definition

We describe an agent system as a triple (A, C, Ω) : a set of agents A , a communication graph C , and an aggregation rule Ω . Figure 1 and Table 1 give the five configurations we evaluate, each paired with the clinical role it instantiates. The taxonomy is the one used for general-domain agent systems (Kim et al., 2026); the clinical instantiations are ours. Only the coordination logic differs across architectures. The item rendering, the answer schema, the role prompts and the decoding parameters are shared.

Independent. N specialists receive the item and answer without seeing one another. Aggregation is majority vote, ties broken by mean self-reported confidence and then by option order. A confidence-weighted variant is logged throughout.

Centralized (attending physician). An orchestrator receives the N specialist opinions on a star topology and is instructed to *audit before aggregating*. It must name the single opinion most likely to be wrong and justify that, then either commit to an answer or re-task one named specialty. A re-tasked specialist re-answers with the orchestrator’s critique in hand. The star topology is preserved, since specialists never see each other. This design gives the panel a single point at which errors can be caught before they reach the output, and it is the architecture responsible for both the largest gain and the largest loss in the study.

Decentralized (MDT discussion). N specialists give first opinions, then exchange opinions on a complete graph for up to three rounds, each free to revise. Rounds stop early only on true unanimity, defined as every agent emitting the same *valid*

option. A panel containing unparsed answers is not unanimous.

Hybrid (tiered referral). A generalist answers first with a stated confidence. Below a frozen threshold θ the case is referred to the N -specialist panel; if that panel has no majority the case escalates to full MDT discussion. θ is calibrated once on a 100-item development slice disjoint from every evaluation set and then frozen.

Panel composition. A cheap router (gpt-4.1-nano, temperature 0) ranks the nine most relevant specialties for each item from a fixed 17-specialty roster; a panel of size N is the first N of that ranking. Panels are therefore *nested*: the $N=9$ panel contains the $N=5$ panel, which contains the $N=3$ panel.

3.2 Nested panels as common random numbers

Every agent’s first opinion is keyed on (item, specialty _{i} , seed, i) and never on N or on the architecture. Consequently the first opinions of a panel of size N are a superset of those for any $N' < N$, and all four multi-agent architectures reuse the same first opinions. This is a deliberate common-random-numbers coupling. It reduces the variance of within-item contrasts between architectures and panel sizes, and it changes the estimand: we measure *accuracy as the same routed panel is extended*, not the expected accuracy of independently redrawn panels at each size. Paired McNemar tests on item-level outcomes remain valid under this coupling; regression models must cluster at the item level rather than treat configurations as independent.

3.3 What we measure

Accuracy alone cannot distinguish a panel that pools complementary evidence from one that repeats itself, so we instrument each episode with five coordination quantities and one derived quantity that the rest of the paper turns on.

We measure five coordination quantities. Their names are established in the agent-scaling literature (Kim et al., 2026), which reports headline values without always giving closed forms; the definitions below are the ones we compute.

- **Turns T :** reasoning–response exchanges; one model call is one turn.

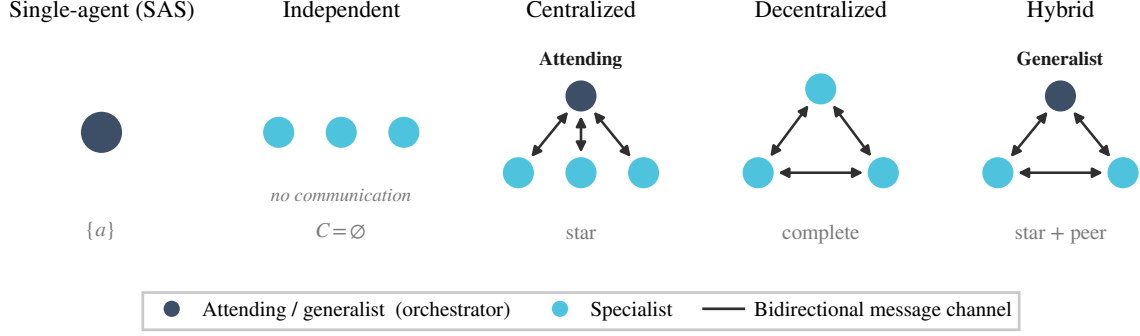


Figure 1: **The five architectures we evaluate**, with the clinical role each instantiates. A is the agent set, C the communication graph, Ω the aggregation rule.

Table 1: Architectures, their coordination structure, and the clinical role each instantiates. n is the panel size, k reasoning iterations, r orchestration rounds, d debate rounds.

Architecture	(A, C, Ω)	Cost	Clinical instantiation
Single-agent (SAS)	$\{a\}$	$O(k)$	One physician, zero-shot or CoT
Independent	$\{a_1..a_n\}$, $C = \emptyset$, synthesis	$O(nk)$	Specialists answer blind; majority vote
Centralized	$+a_{\text{orch}}$, star, hierarchical	$O(rnk)$	Attending physician audits, then decides
Decentralized	$\{a_1..a_n\}$, complete, consensus	$O(dnk)$	Multidisciplinary team discussion
Hybrid	$+a_{\text{orch}}$, star + peer, both	$O(rnk+pn)$	Tiered referral: triage $\rightarrow$ panel $\rightarrow$ MDT

- **Message density c :** peer-opinion exposures per turn. Independent has $C = \emptyset$ and therefore $c = 0$ by construction; Centralized exposes opinions only to the orchestrator; Decentralized and Hybrid expose every agent to every other each round.
- **Redundancy R :** mean pairwise Jaccard overlap of rationale tokens among first opinions. High R means the panel is restating one another rather than contributing independent information.
- **Overhead $O\%$:** token cost relative to the single-agent baseline on the same items, $(\text{tok}_{\text{MAS}} - \text{tok}_{\text{SAS}})/\text{tok}_{\text{SAS}}$.
- **Coordination efficiency E_c :** accuracy divided by the token-cost ratio to the single-agent baseline.
- **Error amplification A_e :** the extent to which individual-agent errors reach the final output without inter-agent verification. We measure that failure directly, in which the panel held the correct option among its first opinions and still emitted a wrong one, normalised by the marginal probability that a single agent is wrong. $A_e > 1$ means the system destroys information it already possessed.

Effective independent opinions. Majority voting converts independent judgements into accuracy; correlated judgements it merely repeats. We therefore summarise a panel not by its size but by how many independent opinions it carries. Writing φ for the mean pairwise error correlation between panel positions, measured as the correlation of their per-item error indicators, the effective number of independent opinions in a panel of N agents is

$$N_{\text{eff}} = \frac{N}{1 + (N-1)\varphi}. \quad (1)$$

At $\varphi = 0$ the N opinions are fully independent and $N_{\text{eff}} = N$; at $\varphi = 1$ they are one opinion copied N times and $N_{\text{eff}} = 1$. This is the standard design-effect correction for clustered observations, and it makes the arithmetic premise of consultation measurable rather than assumed. Reporting N_{eff} alongside N turns out to explain why panel size predicts so little (§4.5).

3.4 Clinical tasks and benchmarks

Three benchmarks, 250 items each, sampled with a fixed seed: MedXpertQA-Text (Zuo et al., 2025) (10 options; stratified by the 12 body systems $\times$ 3 task types), MedQA-USMLE (Jin et al., 2021) (4 options), and the contamination-screened hard split of MedAgentsBench (Tang et al., 2025) (894 items excluding the medqa_5options subset; the

first 250 after stratified shuffling). The primary endpoint is exact match against the dataset’s own gold option key. No LLM judge appears anywhere in the evaluation path.

Every item is additionally tagged easy / medium / hard by the pass rate of three mid-tier models at $k=5$ samples. On MedXpertQA the strata are heavily skewed hard (13 / 62 / 125 on the pilot sample), which is the intended regime.

4 Experiments and results

4.1 Setup

LLMs and capability scaling. Three OpenAI models form the capability ladder the architecture grid is run on, measured rather than assumed (Table 9; the screening probe over eight candidates is Table 11). The ladder is monotone on both benchmarks and spans single-doctor accuracy from 25% to 55% on MedXpertQA, straddling the previously reported 45% threshold from both sides. Four further models—Gemini 3.5 Flash Lite and Gemini 3.7 Flash from Google, Claude Haiku 4.5 and Claude Sonnet 5 from Anthropic—are run on the same items and the same 27-cell grid. Six more—deepseek-v4-flash and deepseek-v4-pro, qwen3.8-flash and qwen3.8-max, glm-5.3-flash and glm-5.3—contribute single-doctor baselines on the same items, which is what the diversity analysis of §4.5 consumes. Thirteen models across six independently trained ecosystems span $I = 34.0$ to 70.9, and the ladder pairs an American and a Chinese model at matched capability at several rungs—deepseek-v4-pro at 59.7 beside gpt-5-mini at 59.2, Gemini 3.5 Flash Lite at 50.3 beside gpt-5-nano at 50.8—so the vendor contrast can be drawn at matched capability rather than confounded with it. A task-grounded capability metric explains more variance than a generic intelligence index (Kim et al., 2026), so we take I to be each model’s mean single-doctor chain-of-thought accuracy across the three benchmarks.

Budget-matched controls. Every configuration is accompanied by three single-model controls. These are zero-shot, chain-of-thought, and **self-consistency at a matched budget**. The self-consistency pool is built from cached $n=8$ sampling calls (the API maximum), so any k is a prefix of the same samples. Nominal cost is charged as $\lceil k/8 \rceil$ prompts plus k completions, which is what

an efficient implementation would actually pay. Panels are compared against the self-consistency cell whose mean per-item *dollar* cost is closest, not the one with a matching sample count. Matching on samples silently favours the panel, because a panel pays the prompt once per agent while self-consistency amortises it across a batch.

Cost accounting is *nominal* throughout. Cached calls are charged at list price, so the economics figures report what a configuration would cost to run, not what our cache made it cost.

Scaling law. We fit the functional form used for general-domain agent systems (Kim et al., 2026),

$$\begin{aligned} P = & \beta_0 + \beta_1 I + \beta_2 I^2 + \beta_3 \log(1+T_{\text{tools}}) \\ & + \beta_4 \log(1+n_a) + \beta_5 \log(1+O\%) \\ & + \beta_6 c + \beta_7 R + \beta_8 E_c + \beta_9 \log(1+A_e) \\ & + \beta_{10} P_{\text{SA}} + \sum_j \beta_j (\text{interactions}), \quad (2) \end{aligned}$$

with interactions $I \times E_c$, $A_e \times P_{\text{SA}}$, $R \times n_a$, $c \times I$ and $P_{\text{SA}} \times \log(1+n_a)$. Our tasks use no tools, so $T_{\text{tools}} = 0$ identically and every term containing it vanishes. We fit the $T=0$ slice, which is the point of the study. Standard errors are clustered by benchmark, and cross-validated R^2 is reported over five folds.

Statistics. Paired McNemar tests with exact binomial p -values on identical item sets, Holm-corrected within each family of comparisons. Wilcoxon 95% confidence intervals for all accuracies. Power was calibrated on a 200-item pilot. Median discordance between configurations is 8.5%, giving a minimum detectable difference of 5.7 percentage points at $n=200$ and 2.6 points at $n=1000$ with 80% power. Configurations that fail for infrastructure reasons (rate limits, timeouts) are recorded with an error status and excluded from accuracy. Scoring them as wrong answers would bias call-heavy architectures downward.

4.2 Main results

Across 420 multi-agent configurations and 144,499 episodes we find that collaboration in medicine is governed by task difficulty, not by panel size. Figure 2 gives the full dose–response surface; Table 2 the underlying numbers.

Which panel you convene does not matter. Taking each architecture at its best panel size in each of the 21 cells gives 126 pairwise architecture contrasts, of which **five are significant** (paired

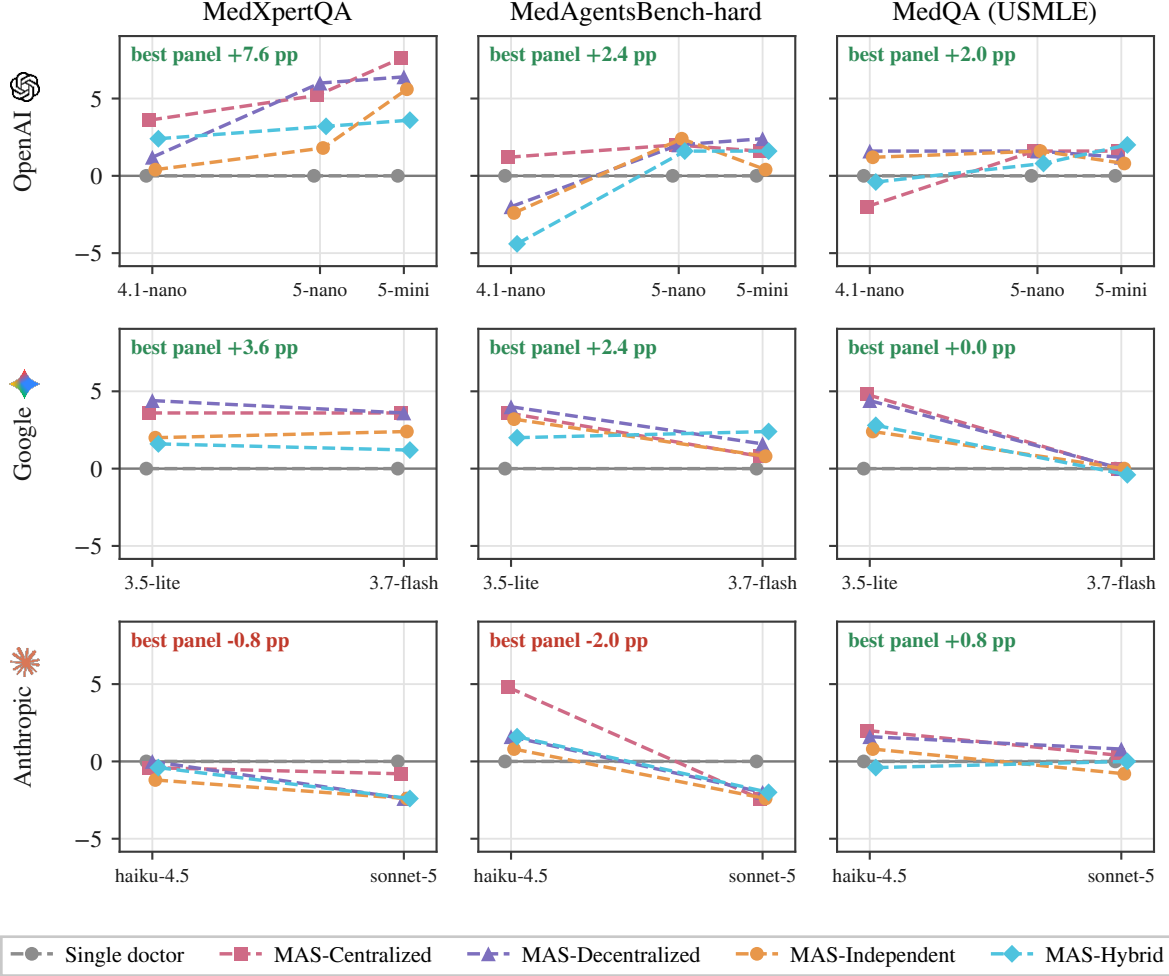


Figure 2: **Agent scaling across model capability and system architecture.** Gain over the single doctor, in percentage points, against the task-grounded capability index I (mean single-doctor accuracy across the three benchmarks); each architecture at its best-performing panel size. Rows are vendors, columns benchmarks, and all nine panels share one vertical scale, so magnitudes are comparable throughout. Plotting absolute accuracy instead would put the three benchmarks on ranges of 15–95% and collapse the architectures onto a single line: the effects being compared span -4.4 to $+7.6$ pp, while capability moves performance by tens of points. Every configuration sits inside a narrow band around the baseline, and the band narrows as the single doctor improves.

McNemar $p < 0.05$, largest difference 5.6 pp). Against the single doctor, 12 of 84 contrasts reach significance. Convening a panel changes the answer on a minority of cells; the topology—star, complete, tiered, or no edges—does not separate at all. Panel size behaves the same way. On gpt-5-nano / MedXpertQA, Independent moves from 31.8% at $N=1$ to 32.6% at $N=9$, which is 0.8 pp for nine times the calls, while Decentralized reaches 38.0% at $N=3$ and stays there. Gains saturate at $N=3$ in every cell where they occur, and no architecture in any cell improves beyond $N=5$.

The best architecture changes with capability tier, within noise. Which architecture posts the highest point estimate does depend on the tier—at

T2 it is Decentralized (38.0%, +6.0 pp) and at T3 Centralized (49.2%, +7.6 pp)—though as above the gap between them is not resolvable at this sample size. Family-dependent architectural preferences have been reported for general-domain agents (Kim et al., 2026), and the pattern here is the medical counterpart. The stronger model exploits an orchestrator’s audit, whereas the weaker model benefits more from peer exchange, whose redundancy compensates for individually unreliable opinions.

4.3 Scaling principles

With accuracy as the response, the fit is degenerate. Fitting that form directly gives $R^2 = 0.997$ (5-fold cross-validated $R^2 = 0.996$), which

Table 2: **Main results.** For each capability tier and benchmark: the single-doctor baseline, then each architecture at its best panel size, and the gain of the best architecture over the single doctor. Bold marks the best architecture in each row. Shaded rows are the four cells whose single-doctor baseline falls between 25 and 50%. The complete 420-configuration grid is released with the code (§E).

	Single doctor	Best multi-agent variant (accuracy % / n_a)				Gain
Tier	doctor	Independ.	Central.	Decentr.	Hybrid	best
<i>MedXpertQA</i> — expert-level reasoning, 10 options						
T1	15.6	16.0/9	19.2 /1	16.8/5	18.0/9	+3.6
T2	32.0	33.8/5	37.2/9	38.0 /7	35.2/5	+6.0
T3	41.6	47.2/5	49.2 /3	48.0/5	45.2/1	+7.6
<i>MedAgentsBench-hard</i> — items where strong models still fail						
T1	20.4	18.0/1	21.6 /7	18.4/5	16.0/9	+1.2
T2	32.4	34.8 /5	34.4/3	34.4/9	34.0/5	+2.4
T3	44.4	44.8/9	46.0/7	46.8 /9	46.0/1	+2.4
<i>MedQA (USMLE)</i> — licensing-exam questions, 4 options						
T1	66.0	67.2/9	64.0/9	67.6 /5	65.6/9	+1.6
T2	88.0	89.6 /5	89.6 /5	89.6 /9	88.8/5	+1.6
T3	91.6	92.4/1	93.2/1	92.8/5	93.6 /1	+2.0

■ shaded rows have a single-doctor baseline between 25 and 50%

is an artefact of our design rather than a result. The response is accuracy and one predictor is the single-doctor baseline P_{SA} ; the same items and the same model underlie both terms and only the coordination structure varies, so P_{SA} alone nearly determines accuracy ($\hat{\beta}_{P_{SA}} = 0.875$, $p < 10^{-4}$). We therefore refit with the collaboration *gain* as the response, which is what the scaling principle is actually about.

Coordination metrics explain the gain that accuracy alone cannot. With $\text{gain} = P - P_{SA}$ as the response, $R^2 = 0.638$ and 5-fold cross-validated $R^2 = 0.415$. Coordination metrics carry that fit with no tool-count term at all. Coefficients ($n = 225$ configurations including the self-consistency control, benchmark-clustered SEs):

term	$\hat{\beta}$	p
$\log(1 + A_e)$	-0.150	< 0.001
R (redundancy)	+0.082	< 0.001
c (message density)	+0.006	< 0.001
$\log(1 + O\%)$	+0.005	0.024
$P_{SA} \times \log(1 + n_a)$	+0.027	0.048
$\log(1 + n_a)$	-0.003	0.701
P_{SA}	-0.113	0.051
P_{SA}^2	+0.037	0.390

Error amplification is the dominant penalty; redundancy and message density carry small positive effects; overhead is not penalised once $T=0$; and neither panel size nor P_{SA} has a main effect.

$T_{\text{tools}} = 0$ throughout, so the $\log(1 + T)$ main effect and the $O\% \times T$ and $E_c \times T$ interactions vanish identically. Coordination metrics remain highly significant without them, which establishes

that the coordination effect is not a re-description of tool overhead. The coefficient on $\log(1 + O\%)$ turns *positive* here. With no tools to coordinate over, spending more tokens on deliberation is mildly beneficial rather than costly.

The decision boundary is not identifiable in the tool-free slice. The published threshold is derived as a coefficient ratio, $P_{SA}^* = \hat{\beta}_4 / \hat{\beta}_{17} \approx 0.063/0.408 = 0.154$ in standardised units, or ≈ 0.45 after denormalisation (Kim et al., 2026). In our fit $\hat{\beta}_{\log(1+n_a)} = +0.0005$ ($p = 0.95$) and $\hat{\beta}_{P_{SA} \times \log(1+n_a)} = +0.0074$ ($p = 0.50$). The ratio, 0.064, rests on two null coefficients and carries no information. This is a statement about functional form, not a failed replication. A linear interaction between baseline and log panel size can express a monotone ceiling. The relationship is not monotone in P_{SA} in the medical data.

Robustness of the fit. Our fit carries I and I^2 with large coefficients of opposite sign (-0.231 and $+0.228$), but bootstrap resampling ($n = 1000$) gives them standard errors of 0.94 and 0.54. With three capability tiers the quadratic is unidentifiable and we do not interpret it. Every other large coefficient is stable: $\log(1 + A_e)$ has bootstrap SE 0.024 and P_{SA} has 0.044.

Residuals pass a normality test (Shapiro–Wilk $p = 0.207$) but fail homoscedasticity decisively (Breusch–Pagan $p = 0.0001$). All standard errors in this paper are therefore heteroscedasticity-robust (HC3), clustered by benchmark where more than two clusters exist. Residual standard error is

$\hat{\sigma} = 0.017$, reflecting that collaboration gains in medicine are small in absolute terms. Regularised alternatives do not improve prediction: Lasso (10-fold CV for λ) retains 9 of 11 predictors at CV $R^2 = 0.382$ and Ridge reaches 0.144, both below the unregularised 0.415, so we retain the full specification.

Nested model comparison isolates where the explanatory power lives (5-fold CV R^2): the full model reaches 0.415; capability alone 0.196; coordination metrics alone 0.093; the single-doctor baseline alone 0.043; and **panel size alone** -0.006 , worse than predicting the mean. Any account of medical consultation that treats “how many agents” as the design variable is fitting noise.

The scaling principle selects an architecture, given a case mix. The fitted model converts into a design rule. For a clinical deployment characterised by its single-doctor baseline P_{SA} , expected collaboration gain is positive only for $P_{SA} \in [25\%, 50\%]$, is maximised by centralized coordination at $n_a = 3$, and is negative below 25%. Three archetypes make this concrete. *Board-style examination triage* ($P_{SA} \approx 0.90$, MedQA-like): use a single model; no architecture we tested returns a significant gain and every panel costs $2\text{--}5\times$ more. *Specialist referral review* ($P_{SA} \approx 0.35$, MedXpertQA-like): use an attending-physician panel at $n_a = 3$; expected gain $+6$ to $+8$ pp at a $6.7\times$ cost premium. *Frontier diagnostic difficulty* ($P_{SA} < 0.25$, MedAgentsBench-like at weak tiers): use a single model and spend the budget on a stronger one; panels lose 2.8 pp here because a panel of unreliable specialists reinforces its own errors.

Error amplification, not panel size, dominates the fit. $\log(1 + A_e)$ carries the largest coefficient of any coordination metric (-0.150 , $p < 10^{-4}$). A_e measures how often a panel held the correct option among its first opinions and still emitted a wrong one. Configurations that discard information they already possess are exactly the configurations that fail to beat the single doctor. The same mechanism has been proposed from the opposite direction for general-domain agents (Kim et al., 2026); our measurement supplies the configuration-level evidence.

$R \times n_a$ is positive ($\hat{\beta} = +0.006$, $p < 0.001$), and resolving R against the ground truth says what it is measuring. Mean rationale overlap is 0.392 for pairs that are both right, 0.351 for pairs that

are both wrong and 0.206 for pairs that split ($t = 119.5$ between the first two, $p < 10^{-300}$). R therefore separates *agreeing* from *disagreeing* by 0.15–0.19 and *agreeing-and-right* from *agreeing-and-wrong* by 0.040: it is a detector of whether two specialists took the same path, and only weakly of whether the path was correct. Its positive coefficient is best read as restatement tracking the items where the model is on familiar ground, not as deliberative verification.

4.4 Coordination efficiency and error dynamics

Coordination is cheap in turns and uninformative in messages. Fitting $T = a(n + 0.5)^b$ gives $a = 0.84$, $b = 0.777$, $R^2 = 0.327$. The exponent is sub-linear, so turns grow more slowly than agents. A clinical panel answering a closed question terminates on unanimity rather than negotiating open-ended tool use. Turn counts by architecture are Decentralized 6.8, Centralized 6.2, Independent 5.0, Hybrid 2.1, against a single-agent 1.4, a range of 1.5 to $4.9\times$. The hard resource ceiling they infer from super-linear growth does not arise in tool-free consultation.

$S = 0.504 - 0.028 \ln(c)$, $R^2 = 0.025$: message density explains almost none of the variance in accuracy. Our Centralized panels run at $c = 0.88$ and Decentralized at $c = 0.60$, well past the $c^* = 0.39$ plateau reported elsewhere, with no accuracy penalty. Once a panel is communicating at all, communicating more neither helps nor hurts.

Coordination failure is structurally impossible without a channel. We classify first-round errors into four categories, following the multi-agent failure taxonomy of Cemri et al. (2025):

	Indep.	Centr.	Decentr.	Hybrid
Logical contradiction	6.3%	6.3%	6.3%	3.4%
Numerical drift	4.5%	4.7%	4.8%	3.1%
Context omission	13.0%	13.3%	13.4%	6.8%
Coordination failure	0.0%	3.0%	1.2%	0.0%

Coordination failure is **0.0%** under Independent coordination, and necessarily so: with $C = \emptyset$ there is no channel on which it could occur. Hybrid is also at 0.0% because its triage gate declines to convene a panel on most items. The magnitudes do not replicate: peer verification is reported to cut logical contradiction from 16.8% to 11.5%, whereas our first-round contradiction rate is flat at 6.3% across communicating and non-communicating architectures. Numerical drift is

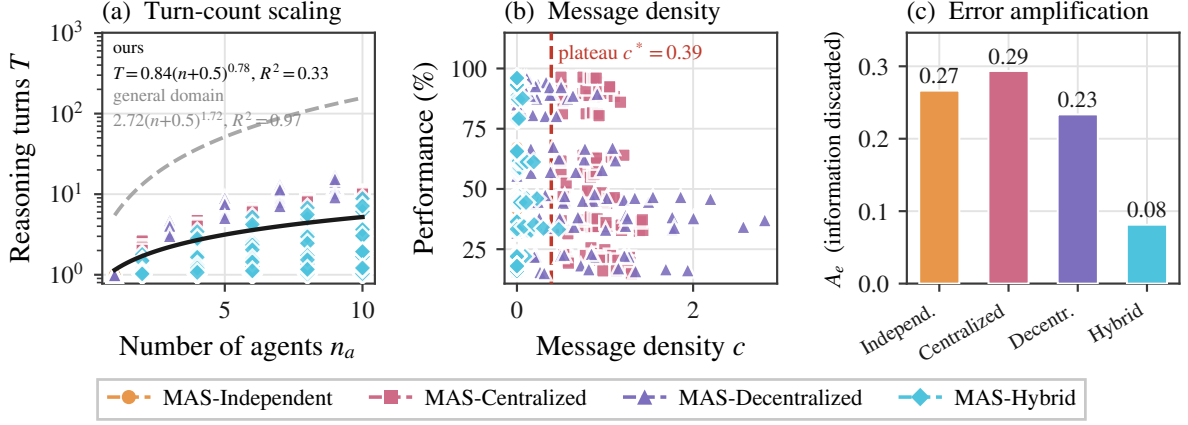


Figure 3: **Coordination dynamics.** (a) Turn count against agent count, with our fit and the published general-domain power law for comparison; medical panels converse an order of magnitude less than that law predicts. (b) Performance against message density, with a reference plateau marked at $c^* = 0.39$; performance is uncorrelated with how much the panel says. (c) Error amplification A_e , the share of information a round discards, by architecture.

an order of magnitude rarer than in tool-using tasks (3–5% against 19–26%), as expected when no arithmetic is cascaded.

Architecture rankings do not generalise across benchmarks. Kendall’s τ between the architecture rankings of any two benchmarks averages +0.067 (pairwise 0.000, +0.200, 0.000; none significant). Leave-one-benchmark-out cross-validation gives $R^2 = -0.015$: a model fitted on two medical benchmarks predicts the third no better than the mean. Coordination principles that transcend task structure in general-domain agentic work do not transcend benchmark in medicine: an architecture chosen on one clinical benchmark carries no guarantee on another.

The architecture that wins on accuracy is the one that costs most. Successes per 1,000 tokens: Hybrid 0.81, Independent 0.41, Decentralized 0.37, self-consistency 0.27, Centralized 0.22. Centralized is $3.7\times$ less token-efficient than the cheapest configuration, and the architecture responsible for every significant gain is the one that costs most per correct answer. Hybrid is the most efficient, because its triage gate declines to escalate most cases and therefore rarely pays for a panel at all.

On MedXpertQA at gpt-5-nano, Decentralized at $N=3$ reaches 38.0% while self-consistency at the closest matched dollar cost reaches 36.0%. Elsewhere the control wins or ties: on MedQA at the same tier, self-consistency at $k=15$ (88.4%) is within noise of the best panel (89.6%) at

Table 3: **Accuracy per dollar.** For each cell, the best configuration of any architecture and panel size: its gain over the single doctor, the additional spend per 1,000 questions, the cost multiple, and the ratio.

Benchmark	Tier	gain	extra \$/1k	$\times$ cost	pp/\$
MedXpertQA	T1	+3.6	0.08	1.9	44.5
MedXpertQA	T2	+6.0	0.70	4.4	8.5
MedXpertQA	T3	+7.6	1.76	3.5	4.3
MedAgentsBench	T2	+2.4	0.28	2.5	8.5
MedAgentsBench	T3	+2.4	7.12	11.5	0.34
MedAgentsBench	T1	+1.2	0.39	6.2	3.1
MedQA	T1	+1.6	0.27	4.2	6.0
MedQA	T2	+1.6	0.49	4.1	3.3
MedQA	T3	+2.0	−0.14	0.8	—

lower cost, and on MedAgentsBench at T1 self-consistency (20.0%) beats every architecture. Where coordination buys anything beyond repetition, it does so narrowly.

What a point of accuracy costs. Expressing each cell’s best configuration as accuracy bought per dollar of additional spend makes the deployment rule concrete (Table 3). The premium ranges from $1.9\times$ to $11.5\times$ the single doctor, and the return on it varies by two orders of magnitude: 8.5 and 4.3 percentage points per additional dollar per thousand questions on the two MedXpertQA cells with the largest gains, against 0.34 on MedAgentsBench at T3, where a +2.4 pp gain costs $11.5\times$. On MedQA at T3 the best configuration is Hybrid at $n_a=1$, which is *cheaper* than the single-doctor baseline because its gate almost never escalates.

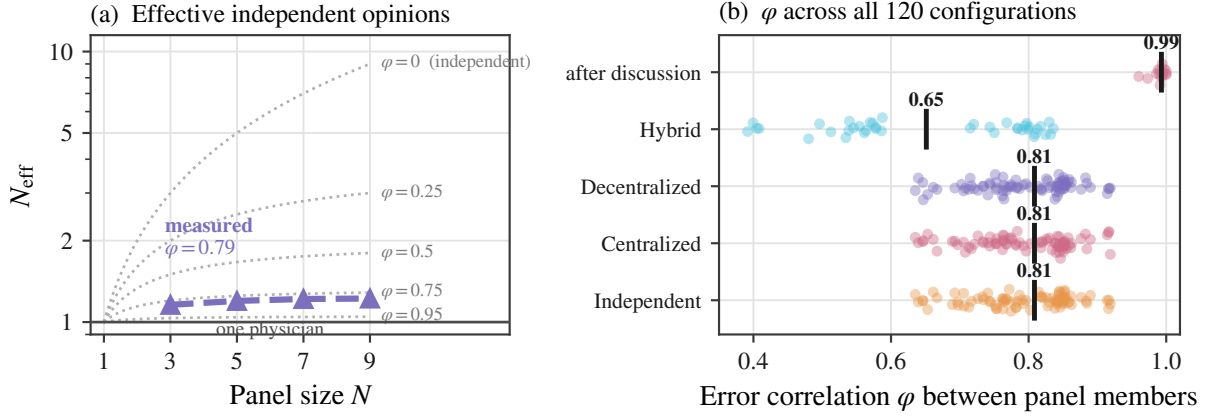


Figure 4: **Adding agents adds almost no information, and discussion removes what is left.** (a) Effective independent opinions $N_{\text{eff}} = N/(1 + (N - 1)\varphi)$ against panel size, on a logarithmic scale. Dotted curves are contours of fixed error correlation φ ; the solid line at $N_{\text{eff}}=1$ is a panel worth one physician. Measured panels track the $\varphi \approx 0.75$ contour, so nine members carry barely more than one opinion. (b) φ for every one of the 292 grid configurations, one point per configuration, with the median marked. The value is a near-constant of the setting rather than an artefact of any one cell, Hybrid routing is the only architecture that lowers it, and peer discussion drives it to 0.99.

Table 4: **Effective independent opinions.** φ is the mean pairwise error correlation between panel positions; $N_{\text{eff}} = N/(1 + (N - 1)\varphi)$. Neither growing the panel nor changing how it is wired moves φ . The shaded row is the single exception: Hybrid’s triage gate declines to convene a panel on most items, so its measured pairs come from the harder subset it does escalate.

Architecture	N	φ	N_{eff}	info. used
<i>Growing one panel: tripling N buys 0.06 of an opinion</i>				
Independent	3	0.796	1.16	39%
Independent	5	0.793	1.20	24%
Independent	9	0.795	1.22	14%
<i>Changing the architecture at $N=9$: no effect either</i>				
Centralized	9	0.796	1.22	14%
Decentralized	9	0.796	1.22	14%
Hybrid	9	0.659	1.43	16%

Table 5: **Clinical safety.** Unanimity is measured on the last round with two or more agents; $P(\text{wr.} | \text{un.})$ is the chance a fully agreeing panel is wrong; the last column is the confidence gap between correct and incorrect answers, the signal a clinician would use to discount an answer. The first row is the single doctor, for reference. Decentralized coordination buys the strongest appearance of agreement and the weakest usable signal at the same time.

Architecture	unan.	$P(\text{wr.} \text{un.})$	conf. gap
Single doctor	—	—	3.1
Independent	69.5%	39.7%	3.8
Centralized	70.6%	38.9%	4.0
Decentralized	98.3%	45.7%	2.4
Hybrid [†]	54.3%	50.6%	2.9

the shaded row agrees on almost every item and is wrong on 46% of them. [†]Hybrid is measured on the 20% of items its triage gate escalates ($n = 1,048$).

4.5 The arithmetic premise of consultation

Majority voting has value only when the votes are independent. Every result above is a consequence of how badly that premise fails.

A nine-member panel carries 1.2 independent opinions. Across the three architectures that convene a full panel on every item at $n_a \geq 3$ — Independent, Centralized and Decentralized—the mean pairwise error correlation is $\varphi = 0.794$, and it is the same to three decimals in each of them separately. (Hybrid is excluded here and reported separately: its triage gate convenes a panel on only 20% of items, so its φ is measured on a harder subset.) By Eq. 1 this leaves $N_{\text{eff}} = 1.16$ at $N=3$,

1.20 at $N=5$ and 1.22 at $N=9$. Tripling the panel adds 0.06 of an independent opinion. Specialty role prompts do not make one model into several physicians; they make it into one physician wearing several labels.

Discussion drives the effective panel size to one. Peer discussion raises φ from 0.775 before the exchange to **0.990** after it (Fig. 4) (paired $t = 26.5$, $p = 2 \times 10^{-13}$). At $\varphi = 0.990$ a nine-agent panel has $N_{\text{eff}} = 1.01$. A multidisciplinary consultation, at the moment it reaches consensus, is informationally equivalent to one physician, and the vote taken at that moment is a vote among copies.

This is the mechanism behind every safety result in §4.6: unanimity rises to 98% because the opinions have merged, not because they have converged on truth; the error rate given unanimity stays at a coin flip because merging does not add information; and confidence rises on wrong answers because N agents agreeing looks like corroboration when it is repetition.

What discussion actually moves. The collapse is not agents falling silent. Across 18,196 (item, panel position) pairs, 38.6% of opinions change between the first round and the last, so deliberation is doing something. Where it takes them is the point. Of the changes, 34.6% go from wrong to right, 22.2% from right to wrong, and 43.2% from one wrong answer to a different wrong one. Two-thirds of all movement ends somewhere other than the correct option. The net effect on individual opinions is mildly positive—872 more corrections than corruptions, or +4.8 pp of positions—and the net effect on the panel is not, because the same exchange that corrects an individual is what makes the others agree with him. Deliberation buys a small improvement in each member and pays for it with the independence that made pooling members worth anything.

The nested model comparison found panel size alone at cross-validated $R^2 = -0.006$. The independence measurement explains why that is not a null result but a necessary one: over $N \in \{3, \dots, 9\}$, N_{eff} moves from 1.16 to 1.22. The regressor that matters barely varies, so the regressor we can control cannot predict. Any design effort spent on how many agents to deploy is spent on the wrong variable.

Putting the right specialist on the case does not help. The arithmetic above says a specialty roster buys little diversity. Medicine offers a more direct test of the same premise. Every MedXpertQA item is labelled with the body system it concerns, and the router ranks specialties by relevance, so at a given panel size some items receive a panel containing the specialty that owns their body system—a neurologist on a nervous-system case—and others do not. If convening the right specialist is what a multidisciplinary panel is for, the first group should gain more.

We compare the *gain over the single doctor* on the same items, which absorbs item difficulty into the baseline. Across three models at $n_a \in \{3, 5\}$ the difference between panels that contain

the matching specialty and panels that do not is -0.05 pp on average, and one of the six contrasts reaches $p < 0.05$, which is what six comparisons produce by chance. A cardiologist on a cardiac case is worth nothing measurable, because the specialty prompt changes what the agent says it is and not what the model knows.

Correlation has a ceiling, and half of the reported decorrelation is that ceiling. Two binary error indicators with unequal marginals cannot reach $\varphi = 1$. Writing p and q for the two members’ error rates, ordered $p \leq q$, the attainable maximum is

$$\varphi_{\max} = \sqrt{\frac{p(1-q)}{q(1-p)}}, \quad (3)$$

which falls as the members’ accuracies diverge. Any panel assembled from models of different strength therefore posts a lower raw φ whether or not its members reason differently, and any analysis that reads raw agreement or overlap as evidence of diversity is partly reading Eq. 3. The correct quantity is the normalised correlation $\varphi/\varphi_{\max}$. Regressing the 45 model pairs in Table 6 on family identity and accuracy gap separates the two. On raw φ the accuracy gap dominates ($\beta = -0.601$, 95% CI $[-1.034, -0.123]$, $R^2 = 0.341$). On $\varphi/\varphi_{\max}$ that coefficient vanishes and changes sign ($+0.156$, 95% CI $[-0.599, +0.994]$, $R^2 = 0.063$), while the family term is unmoved ($+0.093 \rightarrow +0.094$, both intervals excluding zero). Mixing capability does not decorrelate errors at all; it only lowers the ceiling the correlation could have reached. Mixing model families decorrelates genuinely, and by a little.

Independence is for sale, and the price is more than it is worth. The constructions in Table 6 order cleanly, and the ordering is the practitioner’s menu (Fig. 5). Specialty prompts leave $\varphi/\varphi_{\max}$ at 0.766. Swapping in genuinely different checkpoints from one family reaches 0.560, and crossing training families inside a single vendor reaches 0.465—a nine-member panel worth 2.6 opinions, not nine. Crossing the vendor boundary, which is where the remaining hope for independence usually sits, does not continue the trend: every cross-vendor pair we can form among six independently trained ecosystems—OpenAI, Google, Anthropic, DeepSeek, Alibaba and Zhipu, 300 pairs over the same 750 items—gives $\varphi/\varphi_{\max} = 0.504$ and $N_{\text{eff}} = 2.29$, statistically indistinguish-

Table 6: **Four sources of panel diversity.** Normalised correlation $\varphi/\varphi_{\max}$ is the quantity that is not confounded by unequal error rates. Rows relax, in turn, what makes two panel members different: a prompt, a checkpoint, a training family (gpt-4o / gpt-4.1 / gpt-5), and finally the vendor—all pairs across six independently trained ecosystems, three American and three Chinese. The last step—the one usually assumed to be the answer—moves nothing, and moves it the wrong way. n is model–benchmark pairs.

Source of diversity	n	φ	$\varphi_{\max}$	$\varphi/\varphi_{\max}$	N_{eff}^9
One model, different specialty prompts	—	0.734	0.958	0.766	1.31
Different checkpoints, one training family	21	0.479	0.854	0.561	1.86
Different families, one vendor	39	0.308	0.663	0.465	2.60
Different vendors (OpenAI/Google/Anthropic/DeepSeek/Alibaba/Zhipu)	300	0.366	0.726	0.504	2.29
<i>fully independent</i>	—	0	1	0	9.00

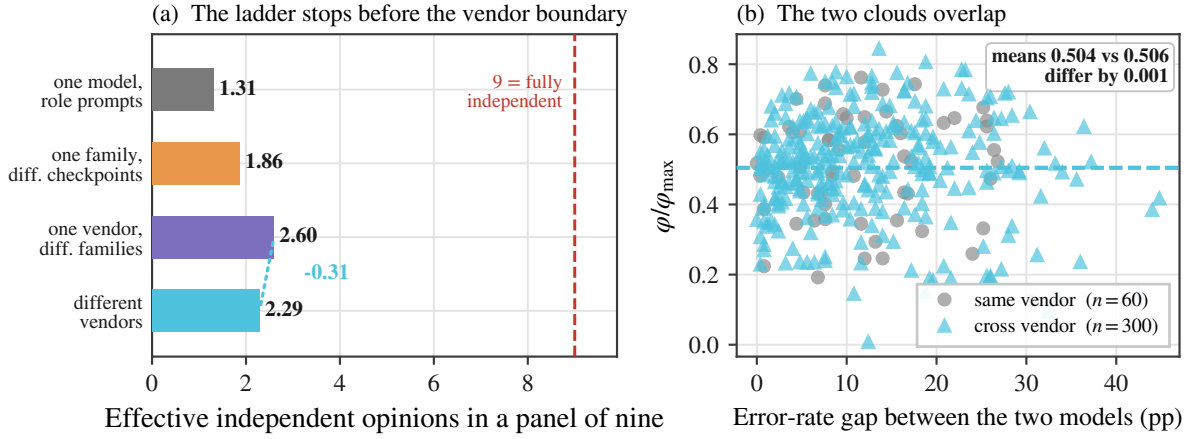


Figure 5: **Independence does not arrive at the vendor boundary.** (a) Effective independent opinions in a nine-member panel under four sources of diversity. Relaxing the prompt, then the checkpoint, then the training family lifts the panel from 1.31 to 2.60; relaxing the vendor moves it -0.31 . (b) The evidence behind that mean: all 360 model–benchmark pairs, normalised correlation against the gap in the two members’ error rates. The same-vendor and cross-vendor clouds occupy the same region and their means differ by 0.001. The rightward drift of both is Eq. 3 at work, which is why the normalised quantity is the one plotted.

able from staying inside one vendor and pointing the wrong way ($\hat{\beta}_{\text{same vendor}} = -0.032$, 95% CI $[-0.088, +0.024]$ on normalised correlation, with the family term at $+0.088$). The comparison is not confounded by capability: the ladder pairs models at matched I on both sides of the boundary—deepseek-v4-pro at 59.7 against gpt-5-mini at 59.2, Gemini 3.5 Flash Lite at 50.3 against gpt-5-nano at 50.8—and restricting to pairs whose error rates differ by less than 5 pp, where $\varphi_{\max}$ averages 0.9 and the mechanical artefact nearly vanishes, leaves cross-vendor pairs at 0.480 against 0.386 within a vendor (2.02 against 2.31 effective opinions), if anything *more* correlated. Two models trained on different continents, on different corpora, under different alignment regimes, fail on the same medical items as reliably as two checkpoints from one lab. The remaining lever is capability spread, and it is a trap. The least correlated pair in the study, gpt-4o-mini

with gpt-5-mini, reaches $\varphi = 0.057$ by way of a 35-point accuracy gap: its members disagree because one of them is much worse. Convened as a panel, a mixed-capability roster reaches $N_{\text{eff}} = 1.72$, the highest we measure, and scores 40.8% on MedXpertQA—indistinguishable from the 41.6% of its strongest member working alone (§F.1). Independence can be bought, but only by seating someone whose errors differ because they are worse, and the panel then performs like the good doctor it diluted.

What independence does not explain. Error correlation accounts for the size and consensus results; it does not by itself predict how much a given panel gains. φ is unrelated to the single-doctor baseline ($r = +0.067$, $p = 0.29$) and weakly related to collaboration gain ($r = -0.297$, $p < 0.001$, $R^2 = 0.09$). It tells us why a panel cannot help much, and why its agreement should

not be read as evidence; how much any particular panel gains is set by what it contains and what aggregation recovers, which is the decomposition of §4.6.

4.6 Why no architecture helps: the answer is in the room

Error correlation explains why averaging fails. It does not explain why architectures that never average—an orchestrator that audits, a debate that revises—fail in the same way. This section closes that gap with a decomposition separating what a panel *contains* from what aggregation *recovers*.

Panels disagree at a human rate, and the disagreement is sterile. Two specialists in the same panel give different answers on 17.3% of items before any exchange (MedXpertQA 24.1%, MedAgentsBench 19.3%, MedQA 7.4%). That is close to the 21% of second opinions that produce a distinctly different diagnosis in a large second-opinion audit (Van Such et al., 2017), so on this measure an AI panel looks about as diverse as a pair of physicians. Resolving the pair against ground truth shows the resemblance is superficial. Across all pairs, 48.5% agree and are right, 38.7% agree and are *both wrong*, 9.6% split with exactly one right, and 7.7% split with both wrong. Of every disagreement, 44.5% is two agents reaching different wrong answers. The panel argues at roughly the human rate, but on two-fifths of items it has already agreed on the wrong diagnosis before the argument starts, and no aggregation rule and no panel size can reach those items.

Ninety-nine percent of the panel’s diversity is a temperature knob. Before asking what aggregation recovers, ask where the diversity comes from. Self-consistency gives the same model the same generic prompt nine times at the same temperature; a panel gives it nine different specialty prompts. Both produce nine answers, so both admit the same oracle measurement. At $n_a=9$ the single doctor scores 52.9%, nine samples of one generalist put a correct answer on the table 62.1% of the time, and nine specialists put one there 62.2% of the time. Of the 9.3 pp a specialty panel appears to offer over one doctor, 9.2 pp—99%—is already available from sampling one generalist repeatedly. The specialty roster contributes +0.06 pp, paired over the 21 model–benchmark cells (95% CI $[-1.96, +2.07]$, $p = 0.95$). The specialty roster is the defining feature of the systems

this design pattern is built on, and the diversity it adds to the room is not measurable beside what a temperature setting already supplies. What that diversity is worth once an architecture has to act on it is a separate measurement, and we make it next.

Role prompting is the panel’s only diversity source until a channel replaces it. The oracle measurement asks what reaches the room; the complementary question is what a panel delivers once an architecture has to act on it. We answer it by rebuilding the panel out of undifferentiated internists: every member receives the same generic system prompt, so panel size is the only thing that varies and the specialty roster is the single removed ingredient. Arms share the model, the temperature, the architectures and the same 250 items (gpt-5-nano, MedXpertQA), and we pair them over the 12 matched (architecture, $N \geq 3$) cells. Against one doctor at 32.0%, the specialist panel gains +4.63 pp. Asking an internist N times supplies +2.77 pp of that (95% CI $[+1.51, +4.03]$, $p = 0.0005$) and the roster supplies +1.87 pp (95% CI $[+0.95, +2.78]$, $p = 0.0009$)—so role prompting is worth just under two points of accuracy, and it is worth them. The split is what identifies the mechanism. It tracks the communication channel: the roster carries 81% of the gain under independent voting, 40% under a central orchestrator and 22% under discussion. With no channel, distinct specialties are the only thing making two panelists disagree, and nine copies of one internist voting by majority are worth +0.50 pp—asking the same doctor again is not a second opinion. Adding a channel manufactures the disagreement that the roster was manufacturing, and substitutes for it rather than compounding with it: the roster alone (independent voting) is worth +2.60 pp and the channel alone (generic panelists, discussion) +4.50 pp, but running both returns +5.80 pp where their sum would predict +7.10 pp. Two routes to disagreement do not buy two panels’ worth of it.

The right answer is in the room far more often than the panel says it. Let the *oracle ceiling* be the accuracy of a rule that always selects a correct first-round opinion whenever any panel member holds one. It upper-bounds every aggregation rule that does not see the label. At $n_a=9$ the single doctor scores 52.9%, the oracle 62.2%, and the best measured architecture 54.2%. The panel holds 9.3 pp of recoverable accuracy, and no

Table 7: **Gain decomposed.** Collaboration gain is headroom times capture rate (Eq. 4). The two factors move in opposite directions, so the product is small throughout. n is configurations.

P_{SA} band	n	headroom	capture κ	gain
< 25%	56	+6.7 pp	−12.0%	−0.27
25–50%	112	+11.4 pp	−13.6%	+0.98
50–70%	44	+6.3 pp	+10.6%	+0.68
> 70%	80	+3.3 pp	+19.6%	+0.68

mechanism we test recovers a tenth of it. Averaged over all configurations at $n_a \geq 3$, an auditing orchestrator captures 4.2% of the headroom, budget-matched self-consistency 3.7% and peer debate 3.7%, while majority vote captures −12.4% and tiered referral −5.8%: both finish below the accuracy of a panel member drawn at random. Selecting the best architecture per cell after the fact, which is not a deployable strategy, reaches 15.0%. The failure is not that a panel of one model has nothing to add. It is that nothing in the panel can tell which member is the one to listen to (Fig. 6).

Gain factors into what the panel holds and what aggregation keeps. Collaboration gain factors exactly as

$$\underbrace{P - P_{SA}}_{\text{gain}} = \underbrace{(P_{\text{oracle}} - P_{SA})}_{\text{headroom}} \times \underbrace{\kappa}_{\text{capture rate}}, \quad (4)$$

where κ is the fraction of the headroom an architecture actually recovers. The factorisation is definitional; what is not definitional is that the two factors move in opposite directions (Table 7). Headroom is largest where the single doctor is weakest and shrinks as the doctor improves, from +11.4 pp at $P_{SA} \in [25, 50)$ to +3.3 above 70%. Capture moves the other way, from −13.6% to +19.6%: the panels with the most to recover are the ones that discard it. Gain is the product, and it is small everywhere—between −0.3 and +1.0 pp across the four bands.

The headroom is item-level routing value, not a better specialty. Two oracles separate what a fixed roster could deliver from what per-item routing could. Always trusting whichever specialty slot turns out best over the whole benchmark—a post-hoc choice no deployment could make in advance—is worth 54.4% against 52.9% for the single doctor. Choosing a correct opinion item by item is worth 62.2%. Of the 9.3 pp of headroom, 1.5 is available from picking a better specialist and

Table 8: **Every arbiter a panel could use.** Discrimination of each first-round signal against whether the panel’s final answer is correct, over all 64,591 panel episodes at $n_a \geq 3$. The spread across signals is 0.034 of AUC.

First-round signal	AUC	Youden J
Lowest member confidence	0.604	0.179
Mean stated confidence	0.603	0.166
Answer entropy	0.591	0.177
Majority share	0.590	0.177
Confidence spread	0.570	0.140

7.8 requires changing which member you believe from case to case. The design question is therefore not which specialists to convene but whether anything can arbitrate between them once convened.

No signal the panel can compute tells it when it is right. Stated confidence is the obvious arbiter, and the natural objection is that a better one exists. We tested every quantity a panel can compute from its own first round: mean stated confidence, the entropy of the answer distribution, the majority share, the lowest member confidence and the spread of confidences. Against the outcome the arbiter needs to predict—whether this panel will answer correctly—all five land between AUC = 0.570 and 0.604 (Table 8), a range of 0.034. The lowest member’s confidence is marginally the best and still separates right from wrong with Youden $J = 0.179$. The problem is not the choice of signal. A panel whose members share 79% of their errors cannot distinguish the case where it agrees because the answer is easy from the case where it agrees because its members share a blind spot, and every statistic computable from its own outputs inherits that ambiguity.

A selector trained on the answers does not beat counting votes. Nothing that reads the panel’s output does better, including a model fitted on the answers. For each of the 64,591 panel episodes we formed one row per distinct candidate answer, with fifteen features computable from the panel’s own first round: vote share, margin over the runner-up, the mean, maximum, minimum and spread of the confidences attached to that candidate, that candidate’s confidence relative to the panel mean, rationale lengths, the number of distinct answers, the entropy of the vote distribution, and panel size. The label is whether the candidate is correct. We trained a logistic regression and a gradient-boosting classifier under five-fold cross-

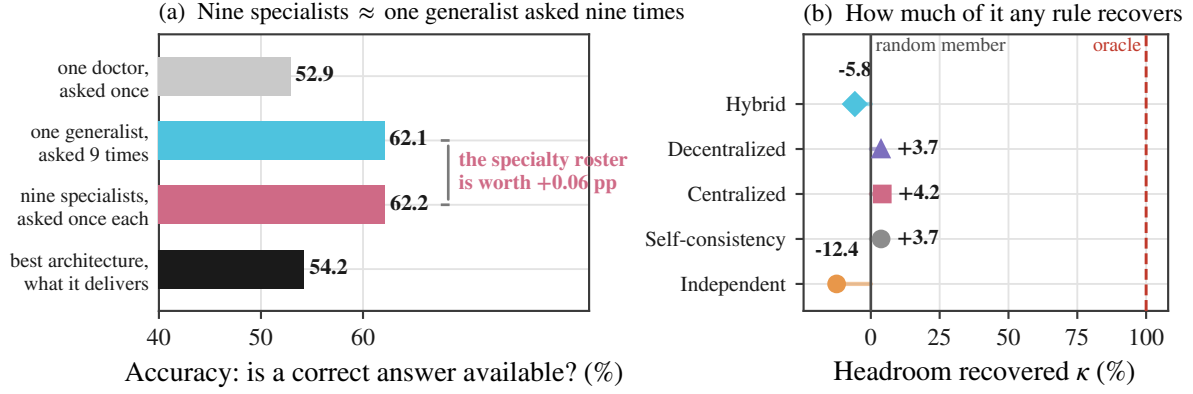


Figure 6: **A panel holds answers it cannot deliver.** (a) How often a correct answer is available at all, at $n_a=9$. Asking one generalist nine times makes one available 62.1% of the time; nine distinct specialists make one available 62.2%. The two middle bars are the comparison the design pattern rests on, and they are the same length: of the 9.3 pp of headroom a specialty panel appears to offer over a single attempt, 99% is sampling and 1% is the roster. The bottom bar is what the best architecture actually returns, 1.4 pp above one doctor. (b) The same gap on a selection scale, where 0 is picking one panel member at random and 100 is an oracle that always picks a correct one. Averaged over all configurations at $n_a \geq 3$, no rule reaches 5, and two fall below zero.

validation grouped by item: fitted on ground truth, tested in the same distribution, with access to everything the panel emits.

Neither beats majority vote. Selecting the highest-scoring candidate gives 52.46% (logistic) and 52.16% (boosted trees) against 52.48% for taking the plurality, while the oracle reaches 59.88%. On the capture scale the two learned selectors score -0.3 and -4.3 : the extra features are noise. The 7.40 pp between plurality and oracle is not merely unexploited by the rules practitioners use; it is unexploited by a model fitted on the answers. Whatever distinguishes the panel’s correct minority from its confident majority is not present in what the panel emits, which is why the productive direction is verification from outside the panel rather than a better rule for reading it.

The one architecture built to select cannot. Eq. 4 says the lever that matters is κ , and κ requires deciding which opinion to trust. Hybrid is the only architecture here that tries: a triage gate escalates to a panel when the first opinion’s stated confidence falls below θ . It escalates on 20% of items overall, and the rate is set almost entirely by which model is behind the gate rather than by which items are hard—76.8% for claude-sonnet-5 on MedXpertQA, 0.0% for gpt-4.1-nano on MedQA, where the gate never once fires in 1,000 items. This is not a poorly tuned threshold. It is the same measurement as §4.7 seen from the other side: a gate can only route on a signal that separates right from wrong,

and stated confidence separates them with Youden $J = 0.166$. Hybrid is cheap and efficient because it is a single model most of the time, and it is a single model most of the time because there is no signal on which it could decide to be anything else.

4.7 Clinical safety: what consultation does to confidence

Accuracy is not the only thing coordination changes. In medicine the more consequential question is what it does to the *signals a clinician would act on*. Three measurements, all pointing the same way.

Self-reported confidence barely discriminates right from wrong. Across all configurations, mean confidence on correct answers exceeds that on wrong answers by only 3.1 points for a single doctor (88.7 vs 85.6; AUC = 0.593; Fig. 7). A nine-member deliberating panel narrows it to 2.4 points (89.5 vs 87.0; AUC = 0.589), and the narrowing is a vendor effect: the three OpenAI models lose 59–78% of the gap, the Google and Anthropic models lose none of it. An auditing orchestrator discriminates best of the five, at AUC = 0.614. Nothing we test reads confidence better than that. This is consistent with the threshold calibration we ran before the main experiments, where the best achievable referral gate on stated confidence reached a Youden J of only 0.166.

Discussion manufactures consensus without manufacturing correctness. Unanimity, de-

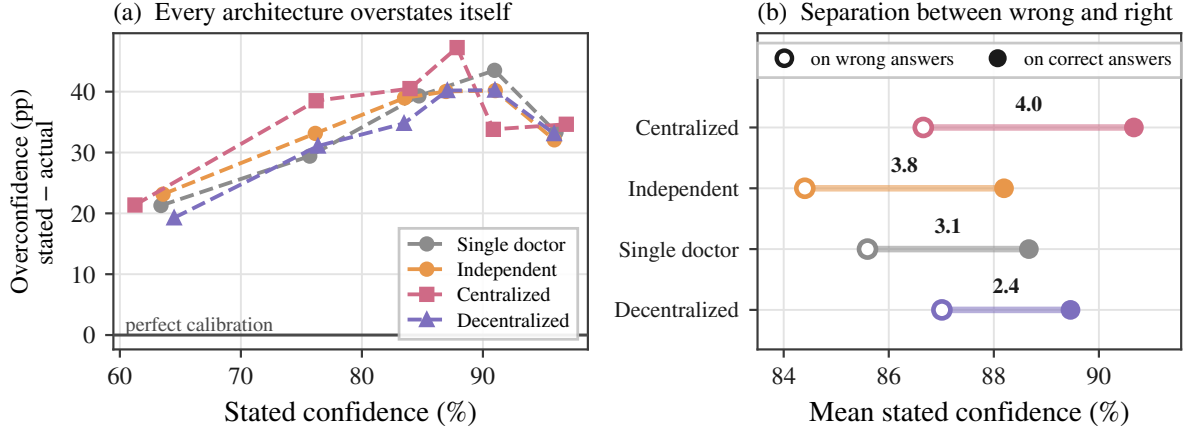


Figure 7: **Consultation degrades the signal a clinician would use to catch its errors.** (a) Overconfidence, defined as stated confidence minus the accuracy actually achieved at that confidence, against stated confidence. Perfect calibration is the solid line at zero; every architecture sits 19–47 pp above it at every confidence level, so panels are not merely uninformative about their own reliability but systematically overconfident. (b) The confidence gap between correct and incorrect answers at $N=9$, which is the quantity a reader would use to discount an answer. Peer discussion cuts it by a quarter, to below the single doctor’s.

fined as every panelist emitting the same valid option in the last round with two or more agents, rises from 69.5% under independent voting to 98.3% under peer discussion at $N=9$. Conditional accuracy does not follow: $P(\text{wrong} \mid \text{unanimous})$ is 39.7% for Independent and 45.7% for Decentralized. A unanimous medical panel is wrong on 40–46% of what it agrees on, and discussion both raises that rate and produces far more unanimous panels. Centralized coordination is the only architecture that lowers it (38.9%), which is the error-containment role the orchestrator is supposed to play.

The most trustworthy-looking output is the least trustworthy. Taken together: consultation drives panels to near-total agreement, leaves the error rate given agreement at 46%, and never makes stated confidence a usable guide to whether the panel is right. The configuration a clinician would most readily act on, a unanimous and high-confidence multidisciplinary recommendation, carries essentially the same error rate as a divided one, and advertises itself more strongly. Accuracy gains of 6–8 pp on the best cells are real, but they are purchased alongside a degradation in the calibration of the signal that would let a clinician catch the remaining errors.

5 Conclusion

A consultation is worth convening only if somebody in the room is right when the others are

wrong, and the room can tell who. Our measurements say that panels of language models have a little of the first and almost none of the second.

The first is scarcer than the design pattern assumes, and it does not come from where the design pattern puts it. Agents prompted into different specialties make the same mistakes ($\varphi = 0.794$), leaving a nine-member panel with 1.22 effective independent opinions, and peer discussion pushes them into near-perfect agreement ($\varphi = 0.990$), leaving a nine-member consultation with 1.01. Of the 9.3 pp that separates one doctor from the best answer anywhere in a nine-member panel, 99% is already available from asking one generalist the same question nine times; the specialty roster—the defining feature of the systems built on this pattern—contributes +0.06 pp, indistinguishable from zero. Independence can be bought, but not cheaply and not usefully: three independently trained model families reach 2.60 effective opinions out of nine and swapping vendor entirely—300 pairs across six ecosystems on two continents—reaches 2.29, and the least correlated panel we can assemble reaches that state by seating a much weaker member and then performs like the stronger member working alone.

The second is the binding constraint. On a scale where picking a panel member at random scores 0 and always picking a correct one scores 100, an auditing orchestrator scores 4.2, peer debate 3.7, and majority voting and tiered referral score below zero, at -12.4 and -5.8 . The panel usually

holds the answer it does not give. The quantity such a system would need—a signal separating a right member from a wrong one—is missing: self-reported confidence separates them with Youden $J = 0.166$, and the one architecture here that tries to route on it convenes a panel on 20% of items and never once on one benchmark–model cell. Collaboration gain factors as available signal times the fraction recovered, and both factors are small.

Three knobs the field is turning, and one it is not. Panel size: over $N \in \{3, \dots, 9\}$ the effective number of opinions moves from 1.16 to 1.22, and panel size alone predicts gain at cross-validated $R^2 = -0.006$. The specialty roster: a temperature setting reproduces 99% of the diversity it buys, and with the roster removed entirely a panel of identical internists still collects 60% of the accuracy the specialist panel delivers. Coordination topology: pairwise architecture contrasts are non-significant, so star, complete, tiered and edgeless are indistinguishable from one another. The knob that is not being turned is verification: every point of headroom the field is trying to reach through richer coordination protocols is already present in the panel and is being discarded for want of a way to recognise it. Where a panel is deployed anyway, the decision should be made by measuring the single-model baseline on the intended case mix first—budget-matched self-consistency matches or beats every architecture we tested at a fraction of the cost—and capability belongs on the agents doing the reasoning rather than the one supervising them, which matters $4.1\times$ more.

The broader caution is about what a consulting system advertises. A unanimous, high-confidence multidisciplinary recommendation is the output a clinician is most likely to act on, and in these experiments it is the output whose reliability is least legible from its own signals: unanimity reaches 98% while the error rate given unanimity stays at 46%, and stated confidence separates the panel’s right answers from its wrong ones at $\text{AUC} \leq 0.62$ under every architecture we tested. Making medical panels agree is easy. Making them *decidable* is the open problem.

6 Limitations and future works

Our tasks are multiple-choice examination items, not clinical encounters. They carry no history tak-

ing, no test ordering, no longitudinal follow-up, and the exact-match endpoint rewards the option-selection component of diagnosis only. The $T = 0$ property that makes the design clean also bounds what it licenses: results here do not extrapolate to tool-using clinical agents. Message density explains almost none of the variance in performance ($R^2 = 0.056$) in this format, and open-ended settings where agents must generate a differential and request evidence give coordination more to carry; we would expect those metrics to matter more there. The selection result does not depend on the format—whatever a panel produces, something has to decide which member to believe.

The diversity ladder spans six independently trained ecosystems—OpenAI, Google, Anthropic, DeepSeek, Alibaba and Zhipu—and measures the vendor boundary rather than assuming anything about it (§4.5); it does not span all of them. Two readings remain open. A vendor whose training corpus or objective departs further from the current consensus than Gemini does from GPT might decorrelate where these two do not, and our grid cannot exclude it. And our benchmarks are static text items, so a vendor difference that showed up only in tool use, retrieval or hedging behaviour would not register here. What the measurement does close is the specific hope that motivated the question—that pooling across the major commercial labs is what makes a panel independent. Between the two largest, at matched capability, it is not.

Panels are nested by construction (§2.2), so the reported curves estimate accuracy as a routed panel is extended, not the expected accuracy of independently redrawn panels at each size. The coupling reduces variance on the architecture contrasts we care about and does not bias within-item paired tests, but a reader interested in the independent-panel estimand should read the curves accordingly.

The specialty router ranks by relevance, so larger panels append progressively less relevant specialists, which entangles “more experts” with “weaker experts” in the specialty-role condition. The generic-internist control separates them (§4.5): with the roster removed and panel size the only variable, the panel still gains $+2.77$ pp, and the roster’s own contribution, $+1.87$ pp, is largest exactly where no communication channel exists. The entanglement is therefore measured rather than assumed, and it does not run in the di-

rection that would manufacture our result—a roster of *decreasing* relevance still adds accuracy.

Difficulty strata are defined by the pass rate of mid-tier models, some of which also appear in the evaluation ladder. This inflates the level of the hard-stratum results but not the within-item slope across panel sizes, which is computed on identical items.

Ethics Statement

This work uses only public benchmark items derived from licensing examinations and published case collections. No patient data, no human subjects and no clinical deployment are involved.

The findings carry a deployment risk that we want to state plainly rather than leave implicit. We show that multi-agent consultation systems manufacture the *appearance* of corroboration without the substance: peer discussion drives unanimity from 70% to 98% while the error rate given unanimity stays at 46%, and stated confidence never separates the panel’s right answers from its wrong ones at better than $AUC = 0.62$. A unanimous, high-confidence multidisciplinary recommendation is simultaneously the output a clinician is most likely to act on and, among the outputs we measure, the one whose reliability is least legible from its own signals. Anyone deploying such a system owes its users an explicit statement that panel agreement is not evidence of correctness, and should not surface consensus or confidence as a trust cue in a clinical interface. The same property makes these systems attractive for producing medical misinformation that reads as authoritative, since apparent expert consensus is cheap to manufacture and, as we show, uninformative.

Our measurements point to three specific mitigations rather than a general caution. First, do not surface unanimity or aggregate confidence as a trust cue: in our data a unanimous panel is wrong 50% of the time, and no statistic a panel can compute about itself—stated confidence, answer entropy, majority share, minimum member confidence—separates its right answers from its wrong ones better than $AUC = 0.60$. An interface that reports “all five specialists agree” is reporting a quantity with almost no diagnostic content while implying the opposite. Second, surface the panel’s *disagreements* instead, and surface them unresolved: on 9.6% of items exactly one member is right, which is where the panel’s whole value

lies, and a resolved consensus destroys precisely that information before the clinician sees it. Third, route verification outward. The one intervention our diversity ladder shows to be large is a judgement from outside the model ecosystem, consistent with the hybrid human–model collectives of Zöller et al. (2025); verification against retrieved evidence or a clinician is a different kind of signal from more agreement among models, and only the different kind changes the arithmetic.

We report the failure mode as a measured property of systems already being built and deployed, not as a novel capability.

References

- Solomon E. Asch. 1956. Studies of independence and conformity: A minority of one against a unanimous majority. *Psychological Monographs*, 70(9).
- Michael L. Barnett, Dhruv Boddupalli, Shantanu Nundy, and David W. Bates. 2019. Comparative accuracy of diagnosis by collective intelligence of multiple physicians vs individual physicians. *JAMA Network Open*, 2(3).
- Bradley Brown, Jordan Juravsky, Ryan Ehrlich, et al. 2024. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv:2407.21787*.
- Mert Cemri et al. 2025. Why do multi-agent llm systems fail? *arXiv*.
- Lingjiao Chen, Jared Quincy Davis, Boris Hanin, Peter Bailis, Ion Stoica, Matei Zaharia, and James Zou. 2024a. Are more llm calls all you need? towards the scaling properties of compound ai systems. In *NeurIPS*.
- Xiangru Chen et al. 2025. A multi-agent conversational framework for medical decision making. *npj Digital Medicine*.
- Xuanzhong Chen, Ye Jin, Xiaohao Mao, Lun Wang, Shuyang Zhang, and Ting Chen. 2024b. RareAgents: Autonomous multi-disciplinary team for rare disease diagnosis and treatment. *arXiv preprint arXiv:2412.12475*.
- Zhen Chen, Zhihao Peng, Xusheng Liang, Cheng Wang, Peigan Liang, Linsheng Zeng, Minjie Ju, and Yixuan Yuan. 2026. MAP: Evaluation and multi-agent enhancement of large language models for inpatient pathways. *npj Health Systems*, 3.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. Improving factuality and reasoning in language models through multiagent debate. In *ICML*.

1437	Zhihao Fan, Jialong Tang, Wei Chen, Siyuan Wang,	Daniel McDuff, and Xin Liu. 2026. Capable lan-	1490
1438	Zhongyu Wei, Jun Xi, Fei Huang, and Jingren Zhou.	guage models can outgrow the benefits of collabo-	1491
1439	2024. AI hospital: Benchmarking large language	ration . <i>Nature Machine Intelligence</i> , 8:1157–1172.	1492
1440	models in a multi-agent medical interaction simula-	Preprint: arXiv:2512.08296 , <i>Towards a Science of</i>	1493
1441	tor. <i>arXiv preprint arXiv:2402.09742</i> .	<i>Scaling Agent Systems</i> .	1494
1442	Rachel S. Goodman, J. Randall Patrinely, Cosby A.	Yubin Kim, Chanwoo Park, Hyewon Jeong, Yik Siu	1495
1443	Stone, et al. 2023. Accuracy and reliability of chat-	Chan, Xuhai Xu, Daniel McDuff, Hyeonhoon Lee,	1496
1444	bot responses to physician questions. <i>JAMA Net-</i>	Marzyeh Ghassemi, Cynthia Breazeal, and Hae Won	1497
1445	<i>work Open</i> , 6(10).	Park. 2024. Mdagents: An adaptive collaboration of	1498
1446	Mark L. Graber. 2013. The incidence of diagnostic	llms for medical decision-making. In <i>NeurIPS</i> .	1499
1447	error in medicine. <i>BMJ Quality & Safety</i> , 22:ii21–	Tiffany H. Kung, Morgan Cheatham, Arielle Mede-	1500
1448	ii27.	nilla, et al. 2023. Performance of chatgpt on usmle.	1501
1449	Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Wein-	<i>PLOS Digital Health</i> , 2(2).	1502
1450	berger. 2017. On calibration of modern neural net-	Ralf H. J. M. Kurvers, Stefan M. Herzog, Ralph Her-	1503
1451	works and llm confidence. <i>ICML</i> .	twig, Jens Krause, Patricia A. Carney, Andy Bogart,	1504
1452	Wolf E. Hautz and Juliane E. Kämmer. 2024. Whether	Giuseppe Argenziano, Iris Zalaudek, and Max Wolf.	1505
1453	two heads are better than one is the wrong question.	2016. Boosting medical diagnostics by pooling in-	1506
1454	<i>Diagnosis</i> .	dependent judgments. <i>PNAS</i> , 113(31):8777–8782.	1507
1455	Jordan Hoffmann, Sebastian Borgeaud, Arthur Men-	Jialing Li, Zhouhong Gu, Yin Cai, and Hongwei Feng.	1508
1456	sch, et al. 2022. Training compute-optimal large lan-	2026. Scaling behavior of single llm-driven multi-	1509
1457	guage models. <i>NeurIPS</i> .	agent systems. <i>arXiv:2606.00655</i> .	1510
1458	Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiawu	Junyou Li, Qin Zhang, Yangbin Yu, Qiang Fu, and De-	1511
1459	Zheng, et al. 2024. Metagpt: Meta programming for	heng Ye. 2024. More agents is all you need. <i>TMLR</i> .	1512
1460	a multi-agent collaborative framework. In <i>ICLR</i> .	Wenliang Li, Rui Yan, Xu Zhang, Li Chen, Hongji Zhu,	1513
1461	Irving L. Janis. 1972. Victims of groupthink: A psy-	Jing Zhao, Junjun Li, Mengru Li, Wei Cao, Zihang	1514
1462	chological study of foreign-policy decisions and fi-	Jiang, Wei Wei, Kun Zhang, and Shaohua Kevin	1515
1463	ascoes. <i>Houghton Mifflin</i> .	Zhou. 2025. MACD: Multi-agent clinical diagno-	1516
1464	Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng,	sis with self-learned knowledge for LLM. <i>arXiv</i>	1517
1465	Hanyi Fang, and Peter Szolovits. 2021. What dis-	<i>preprint arXiv:2509.20067</i> .	1518
1466	ease does this patient have? a large-scale open do-	Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang,	1519
1467	main question answering dataset from medical ex-	Yan Wang, Rui Wang, Yujiu Yang, Zhaopeng Tu,	1520
1468	ams. <i>Applied Sciences</i> , 11(14).	and Shuming Shi. 2024. Encouraging divergent	1521
1469	Saurav Kadavath, Tom Conerly, Amanda Askell, et al.	thinking in large language models through multi-	1522
1470	2022. Language models (mostly) know what they	agent debate. In <i>EMNLP</i> .	1523
1471	know. In <i>arXiv:2207.05221</i> .	Philip R. Liu, Sparsh Bansal, Jimmy Dinh, Aditya	1524
1472	Juliane E. Kämmer, Wolf E. Hautz, Stefan M. Her-	Pawar, Ramani Satishkumar, Shail Desai, Neeraj	1525
1473	zog, Olga Kunina-Habenicht, and Ralf H. J. M.	Gupta, Xin Wang, and Shu Hu. 2025. MedChat:	1526
1474	Kurvers. 2017. The potential of collective intel-	A multi-agent framework for multimodal diagno-	1527
1475	ligence in emergency medicine: Pooling medical	sis with large language models. <i>arXiv preprint</i>	1528
1476	students’ independent decisions improves diagnostic	<i>arXiv:2506.07400</i> .	1529
1477	performance. <i>Medical Decision Making</i> , 37(6):715–	Y. Liu, Zunamys I. Carrero, X. Jiang, et al. 2026.	1530
1478	724.	Benchmarking large language model-based agent	1531
1479	Jared Kaplan, Sam McCandlish, Tom Henighan,	systems for clinical decision tasks . <i>npj Digital</i>	1532
1480	Tom B. Brown, et al. 2020. Scaling laws for neu-	<i>Medicine</i> .	1533
1481	ral language models. <i>arXiv:2001.08361</i> .	Siqi Ma, Jiajie Huang, Fan Zhang, Yue Shen, Jin-	1534
1482	Elliot Kim, Avi Garg, Kenny Peng, and Nikhil Garg.	lin Wu, Guohui Fan, Zhu Zhang, and Zelin Zang.	1535
1483	2025. Correlated errors in large language models.	2025. MedLA: A logic-driven multi-agent frame-	1536
1484	In <i>ICML</i> .	work for complex medical reasoning with large lan-	1537
1485	Yubin Kim, Ken Gu, Chanwoo Park, Chunjong Park,	guage models. <i>arXiv preprint arXiv:2509.23725</i> .	1538
1486	Samuel Schmidgall, A. Ali Heydari, Yao Yan, Zhi-	Daniel McDuff, Mike Schaekermann, Tao Tu, Anil	1539
1487	han Zhang, Yuchen Zhuang, Yun Liu, Mark Mal-	Palepu, et al. 2025. Towards accurate differen-	1540
1488	hotra, Paul Pu Liang, Hae Won Park, Yuzhe Yang,	tial diagnosis with large language models. <i>Nature</i> ,	1541
1489	Xuhai Xu, Yilun Du, Shwetak Patel, Tim Althoff,	642:451–457.	1542

1543	Pranav Pushkar Mishra, Mohammad Arvan, and Mo-	Xiangru Tang et al. 2025. Medagentsbench: Bench-	1598
1544	han Zalake. 2026. Teammedagents: Pareto-efficient	marking thinking models and agent frameworks	1599
1545	multi-agent medical reasoning through teamwork	for complex medical reasoning. <i>Patterns /</i>	1600
1546	theory. <i>arXiv:2508.08115</i> .	<i>arXiv:2503.07459</i> .	1601
1547	Harsha Nori, Nicholas King, Scott Mayer McKin-	Tao Tu, Anil Palepu, Mike Schaeckermann, Khaled	1602
1548	ney, Dean Carignan, and Eric Horvitz. 2023. Ca-	Saab, et al. 2025. Towards conversational diagnostic	1603
1549	pabilities of gpt-4 on medical challenge problems.	artificial intelligence. <i>Nature</i> , 642:442–450.	1604
1550	<i>arXiv:2303.13375</i> .		
1551	Harsha Nori, Naoto Usuyama, Nicholas King, et al.	Monica Van Such, Robert Lohr, Thomas Beckman,	1605
1552	2024. From medprompt to o1: Exploration of	and James M. Naessens. 2017. Extent of diagnos-	1606
1553	run-time strategies for medical challenge problems.	tic agreement among medical referrals . <i>Journal of</i>	1607
1554	<i>arXiv:2411.03590</i> .	<i>Evaluation in Clinical Practice</i> , 23(4):870–874.	1608
1555	Jesutofunmi A. Omiye, Jenna C. Lester, Simon	Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc	1609
1556	Spichak, Veronica Rotemberg, and Roxana	Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery,	1610
1557	Daneshjou. 2023. Large language models propagate	and Denny Zhou. 2023. Self-consistency improves	1611
1558	race-based medicine. <i>npj Digital Medicine</i> , 6.	chain of thought reasoning in language models. In	1612
		<i>ICLR</i> .	1613
1559	Emma Radcliffe et al. 2019. Multidisciplinary tumour	Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten	1614
1560	boards and their impact on cancer care. <i>The Lancet</i>	Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le,	1615
1561	<i>Oncology</i> .	and Denny Zhou. 2022. Chain-of-thought prompt-	1616
1562	Akshat Sanghvi, Naren Akash, Raza Imam, Amit	ing elicits reasoning in large language models. In	1617
1563	Sharma, and Mohit Jain. 2026. MeDxAgent: Multi-	<i>NeurIPS</i> .	1618
1564	agent consultation for interactive medical diagnosis.	Max Wolf, Jens Krause, Patricia A. Carney, Andy Bog-	1619
1565	<i>arXiv preprint arXiv:2606.03416</i> .	art, and Ralf H. J. M. Kurvers. 2015. Collective	1620
1566	Daling Shi, Tyler Shugg, Michael T. Eadon, Jing	intelligence meets medical decision-making. <i>PLOS</i>	1621
1567	Su, Yijiang Chen, and Qianqian Song. 2026.	<i>ONE</i> , 10(8).	1622
1568	PHO-Agents: A large language model-powered	Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu,	1623
1569	multi-agent system for predicting health outcomes .	et al. 2024. Autogen: Enabling next-gen llm appli-	1624
1570	<i>medRxiv preprint</i> .	cations via multi-agent conversation. In <i>COLM</i> .	1625
1571	Hardeep Singh, Ashley N. D. Meyer, and Eric J.	Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie	1626
1572	Thomas. 2014. The frequency of diagnostic errors	Fu, Junxian He, and Bryan Hooi. 2024. Can llms	1627
1573	in outpatient care. <i>BMJ Quality & Safety</i> , 23(9).	express their uncertainty? an empirical evaluation	1628
1574	Karan Singhal, Shekoofeh Azizi, Tao Tu, S. Sara Mah-	of confidence elicitation in llms. In <i>ICLR</i> .	1629
1575	davi, Jason Wei, Hyung Won Chung, et al. 2023.	Yingxuan Yang, Chengrui Qu, Muning Wen, Laixi	1630
1576	Large language models encode clinical knowledge.	Shi, Ying Wen, Weinan Zhang, Adam Wierman, and	1631
1577	<i>Nature</i> , 620:172–180.	Shangding Gu. 2026. Understanding agent scal-	1632
1578	Karan Singhal, Tao Tu, Juraj Gottweis, Rory Sayres,	ing in llm-based multi-agent systems via diversity.	1633
1579	et al. 2025. Toward expert-level medical ques-	<i>arXiv:2602.03794</i> .	1634
1580	tion answering with large language models. <i>Nature</i>	Grace Chang Yuan, Xiaoman Zhang, Sung Eun Kim,	1635
1581	<i>Medicine</i> , 31:943–950.	and Pranav Rajpurkar. 2026. Do mixed-vendor	1636
1582	Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Ku-	multi-agent llms improve clinical diagnosis? <i>arXiv</i>	1637
1583	mar. 2024. Scaling llm test-time compute optimally	<i>preprint arXiv:2603.04421</i> .	1638
1584	can be more effective than scaling model parameters.	Travis Zack, Eric Lehman, Mirac Suzgun, et al. 2024.	1639
1585	<i>arXiv:2408.03314</i> .	Assessing the potential of gpt-4 to perpetuate racial	1640
1586	James Surowiecki. 2004. <i>The Wisdom of Crowds</i> . Dou-	and gender biases in health care. <i>The Lancet Digital</i>	1641
1587	bleday.	<i>Health</i> , 6(1).	1642
1588	Ivan Sviridov, Amina Miftakhova, Artemiy	Jintian Zhang, Xin Xu, Ningyu Zhang, Shumin Deng,	1643
1589	Tereshchenko, Galina Zubkova, Pavel Blinov,	Chuanqi Tan, and Huajun Chen. 2024. Exploring	1644
1590	and Andrey Savchenko. 2025. 3MDBench: Med-	collaboration mechanisms for llm agents. <i>ACL</i> .	1645
1591	ical multimodal multi-agent dialogue benchmark.	Huiya Zhao, Yinghao Zhu, Zixiang Wang, Yasha Wang,	1646
1592	<i>arXiv preprint arXiv:2504.13861</i> .	Junyi Gao, and Liantao Ma. 2025. Confagents: A	1647
1593	Xiangru Tang, Anni Zou, Zhuosheng Zhang, Ziming	conformal-guided multi-agent framework for cost-	1648
1594	Li, Yilun Zhao, Xingyao Zhang, Arman Cohan, and	efficient medical diagnosis. <i>arXiv:2508.04915</i> .	1649
1595	Mark Gerstein. 2024. Medagents: Large language		
1596	models as collaborators for zero-shot medical rea-		
1597	soning. In <i>Findings of ACL</i> .		

Yucheng Zhou, Lingran Song, and Jianbing Shen. 2025. MAM: Modular multi-agent framework for multi-modal medical diagnosis via role-specialized collaboration. *arXiv preprint arXiv:2506.19835*.

Xiaochen Zhu, Caiqi Zhang, Yizhou Chi, Tom Stafford, Nigel Collier, and Andreas Vlachos. 2026. Demystifying multi-agent debate: The role of confidence and diversity. *arXiv:2601.19921*.

Yinghao Zhu, Ziyi He, Haoran Hu, Xiaochen Zheng, Xichen Zhang, Zixiang Wang, Junyi Gao, Liantao Ma, and Lequan Yu. 2025. Medagent-board: Benchmarking multi-agent collaboration with conventional methods for diverse medical tasks. *arXiv:2505.12371*.

Nikolas Zöller, Julian Berger, Irving Lin, Nathan Fu, Jayanth Komarneni, Gioele Barabucci, Kyle Laskowski, Victor Shia, Benjamin Harack, Eugene A. Chu, Vito Trianni, Ralf H. J. M. Kurvers, and Stefan M. Herzog. 2025. *Human-AI collectives most accurately diagnose clinical vignettes. Proceedings of the National Academy of Sciences*, 122(24).

Yuxin Zuo, Shang Qu, Yifei Li, Zhangren Chen, Xuekai Zhu, Ermo Hua, Kaiyan Zhang, Ning Ding, and Bowen Zhou. 2025. Medxpertqa: Benchmarking expert-level medical reasoning and understanding. In *ICML*.

A Model capability index

Prior work scores models on a published Intelligence Index and reports that a *task-grounded* capability metric explains more variance ($R^2 = 0.413$) than the generic one ($R^2 = 0.373$). We therefore use the task-grounded form throughout: I is a model’s mean single-doctor chain-of-thought accuracy across the three benchmarks. Table 9 gives the components. The three tiers span $I = 34.0$ to 59.2 , and their single-benchmark baselines span 15.6% to 91.6% , crossing the 45% region from both sides. That crossing is what makes the capability-ceiling test possible at all.

B Domain complexity

B.1 Complexity metric construction

Domain complexity $D \in [0, 1]$ is the arithmetic mean of three complementary measures (Kim et al., 2026): the **performance ceiling** $1 - p_{\max}$, where $p_{\max}$ is the highest performance any evaluated system reaches; the **coefficient of variation** σ/μ of performance across all configurations, a scale-invariant measure of relative spread; and the **best-model baseline** $1 - p_{\text{best}}$, where p_{best} is the strongest single-model performance on the dataset.

B.2 Domain characterisation

Table 10 gives the components. MedQA is a low-complexity domain ($D = 0.097$); MedXpertQA and MedAgentsBench-hard are high-complexity and nearly indistinguishable ($D = 0.489$ and 0.478).

B.3 The critical threshold does not transfer

A critical complexity threshold at $D \approx 0.40$ has been reported for general-domain agents (Kim et al., 2026): below it multi-agent architectures yield net positive returns, above it coordination overhead consumes resources otherwise spent on reasoning. Our data do not support a benchmark-level threshold of any value. MedXpertQA ($D = 0.489$) and MedAgentsBench-hard ($D = 0.478$) differ by 0.011 in complexity yet by 3.5 pp in mean multi-agent gain ($+3.09$ pp versus -0.37 pp), giving opposite signs at effectively the same D . A benchmark-level statistic cannot separate them because the quantity that does separate them varies *within* each benchmark: on MedAgentsBench-hard the per-tier gains run -2.78 , $+1.02$, $+0.66$ pp as P_{SA} moves from 20.4% through 32.4% to 44.4% . Domain complexity is the right idea at the wrong granularity for medicine. The configuration-level baseline P_{SA} is what carries the signal.

C Datasets

MedXpertQA-Text (Zuo et al., 2025): 2,450 expert-level items, ten options each, tagged by 12 body systems and 3 task types. We draw a stratified sample interleaved across (body system $\times$ task) so that any prefix is approximately balanced, then take the first 250. **MedQA-USMLE** (Jin et al., 2021): 1,273 four-option test items; randomly shuffled with a fixed seed, first 250. **MedAgentsBench-hard** (Tang et al., 2025): the contamination-screened hard split, 894 items after excluding the medqa_5options subset, stratified-interleaved across the ten source subsets, first 250. All samples are nested prefixes, so a larger sample re-uses every call of a smaller one.

Item difficulty is additionally tagged easy/medium/hard by the pass rate of three mid-tier models at $k=5$; on the MedXpertQA sample this yields 13/62/125 items.

Table 9: Capability index and its components (single-doctor CoT accuracy, %).

Model	MedXpertQA	MedAgentsBench	MedQA	I (mean)	Setting
<i>OpenAI — the capability ladder the grid is run on</i>					
 gpt-4.1-nano	15.6	20.4	66.0	34.0	—
 gpt-5-nano	32.0	32.4	88.0	50.8	effort=low
 gpt-5-mini	41.6	44.4	91.6	59.2	effort=low
<i>Google — a second vendor, for the diversity ladder</i>					
 gemini-3.5-flash-lite	34.8	31.2	84.8	50.3	—
 gemini-3.7-flash	60.4	56.4	95.8	70.9	—
<i>Anthropic — a third vendor, for the diversity ladder</i>					
 claude-haiku-4.5	23.2	20.8	79.6	41.2	—
 claude-sonnet-5	49.6	47.6	94.0	63.7	—
<i>DeepSeek / Alibaba / Zhipu — independently trained ecosystems, single-doctor baselines only</i>					
 deepseek-v4-flash	37.2	39.6	88.0	54.9	—
 deepseek-v4-pro	44.8	42.0	92.4	59.7	—
 qwen3.8-flash	34.4	32.8	88.8	52.0	—
 qwen3.8-max	42.0	40.0	88.8	56.9	—
 glm-5.3-flash	47.6	45.6	93.2	62.1	—
 glm-5.3	42.8	46.4	93.6	60.9	—

Capability-matched cross-vendor pairs:

gemini-3.5-flash-lite (50.3) with gpt-5-nano (50.8), and
deepseek-v4-pro (59.7) with gpt-5-mini (59.2) across the Pacific.

Table 10: Domain complexity of the three medical benchmarks.

Benchmark	$1 - p_{\max}$	σ/μ	$1 - p_{\text{best}}$	D	mean MAS gain
MedXpertQA	0.360	0.379	0.396	0.378	+1.27 pp
MedAgentsBench-hard	0.412	0.350	0.436	0.399	+0.01 pp
MedQA (USMLE)	0.036	0.114	0.036	0.062	+0.37 pp

D Implementation details

D.1 Technical infrastructure

All models are accessed through one OpenAI-compatible client with a content-addressed disk cache, exponential backoff with rate-limit-aware retry, a per-call JSONL ledger recording tokens, latency and cost, and a cross-process spend journal guarded by an OS file lock. Every LLM call is cached on the full request (model, system, messages, temperature, seed, max tokens, effort, JSON mode, tag), so a re-run is a replay.

D.2 Agent configuration

Single agents answer in one call. Independent panels deploy N specialists with synthesis-only aggregation. Centralized systems use one orchestrator over N sub-agents with a maximum of 2 orchestration rounds and one re-task per round. Decentralized systems run N agents through up to 3 discussion rounds, stopping early only on true unanimity. Hybrid systems combine a generalist triage gate with the specialist panel and escalate to full discussion when the panel has no ma-

jority. Agent temperature is 0.7 within panels, 0.3 for single-doctor baselines and for orchestrator/moderator turns; reasoning models are called at effort=low with a minimum completion budget of 1,600 tokens, since reasoning tokens are drawn from the same budget and a smaller cap returns empty output.

D.3 Prompt compilation

Every agent receives the same rendered item (stem, then options as (A) ...) and the same answer schema: JSON with answer, confidence (0–100) and reason (at most 45 words). Only the system prompt differs, naming the specialty. Peer context in communicating architectures is capped at a fixed 900-character budget, balanced-truncated across peers and *independent of N* , so a panel-size effect cannot be a context-length effect. Peer ordering is deterministically rotated per (item, round) so that the first-listed speaker is decorrelated from the top-ranked specialty.

Table 11: **Model selection probe.** All eight candidates screened on 40 items per benchmark with single-doctor CoT before the grid was run. Three OpenAI checkpoints were chosen as the capability ladder because they are monotone and straddle the 45% threshold; two Google models were added later for the vendor contrast. Full 250-item accuracies for the five selected models are in Table 9.

Vendor	Model	Setting	Role	MedXpertQA	MedQA	USD / 1k items
<i>Selected</i>						
OpenAI	gpt-4.1-nano	—	tier T1	25.0%	75.0%	0.09
OpenAI	gpt-5-nano	effort=low	tier T2	37.5%	95.0%	0.17
OpenAI	gpt-5-mini	effort=low	tier T3	55.0%	97.5%	0.64
Google	gemini-3.5-flash-lite	—	vendor contrast	30.0%	95.0%	0.09
Google	gemini-3.7-flash	—	vendor contrast	80.0%	97.5%	0.41
<i>Screened, not selected — used only for the pairwise φ analysis</i>						
OpenAI	gpt-4o-mini	—	—	20.0%	87.5%	0.13
OpenAI	gpt-4.1-mini	—	—	32.5%	85.0%	0.41
OpenAI	gpt-5.4-nano	effort=low	—	47.5%	95.0%	0.14

D.4 Evaluation methodology

Sample sizes. 250 items per benchmark, 420 multi-agent configurations, 144,499 episodes. **Controls.** All architectures share the item rendering, answer schema, role-prompt template and decoding parameters; only coordination logic differs. Model weights are frozen. The primary endpoint is exact match against the dataset’s own gold option key. No LLM judge appears anywhere in the evaluation path. Episodes that fail for infrastructure reasons carry an error status and are excluded from accuracy rather than scored as wrong, which would bias call-heavy architectures downward. Cost accounting is nominal: cached calls are charged at list price, so reported economics describe what a configuration would cost to run.

E Complete results grid

The per-configuration grid — 420 multi-agent configurations and their single-model controls, with accuracy, confidence interval and cost for each — is released with the code rather than printed here, together with the raw episodes it is computed from.

F Additional results

F.1 Where capability belongs inside a panel

Specialist capability matters four times more than the orchestrator’s. Replacing the whole panel with weak models costs 34.4 pp (49.2% to 14.8%). Splitting that substitution isolates where the loss comes from. Downgrading only the *orchestrator*, keeping strong specialists, costs 3.6 pp, so the panel retains 90% of the homogeneous-high span. Downgrading only the *specialists*, keeping a strong orchestrator, costs 14.8 pp, retaining 57%.

A strong attending physician recovers more than half of what a weak panel loses, but a weak attending physician barely dents a strong one. The asymmetry is $4.1\times$. In a centralized clinical panel, capability should therefore be spent on the agents doing the reasoning rather than on the agent doing the auditing.

An orchestrator too weak to audit is worse than no orchestrator at all. The largest single degradation anywhere in this study is a Centralized panel with a weak model on MedQA at $P_{SA} = 66.0\%$: -13.60 pp, $p = 10^{-5}$. Inspection of those episodes shows the mechanism directly. The orchestrator overrides a correct specialist majority. This is the error-amplification term in its most extreme form. It is also the counterpart to the previous result. The audit is what makes centralized coordination work, so an auditor that cannot audit converts the architecture’s strength into its worst liability.

Mixed-capability panels land between their components, with no emergent benefit. A panel of one strong, one mid and one weak specialist reaches 40.8% under Centralized and 44.0% under Decentralized, against 49.2% and 47.6% for homogeneous-high. Both sit below the strong homogeneous panel and well above the weak one, roughly where a capability-weighted average would put them. We find no medical analogue of the heterogeneous-mixing advantage reported for one model family (Kim et al., 2026), in which a low-capability orchestrator with high-capability sub-agents outperformed a homogeneous high-capability panel by 31%. In our data that configuration is the second-best Centralized cell but still 7% below homogeneous-high, directionally

matching what they observe for OpenAI models, where heterogeneous centralized configurations degrade.

F.2 Where in medicine collaboration pays

Collaboration helps diagnosis most, and it is the only task type significant at both tiers. MedXpertQA labels each item as Diagnosis, Treatment, or Basic Science. Splitting by label, the best panel gains +9.5 pp on Diagnosis at gpt-5-nano ($p = 0.008$) and +10.7 pp at gpt-5-mini ($p = 0.004$), against +7.1 and +8.3 pp on Treatment and +7.3 and +8.5 pp on Basic Science. Diagnosis is the only category significant at both tiers. This is the pattern clinical intuition predicts: differential diagnosis is the medical task that most resembles pooling independent perspectives, and it is where pooling pays.

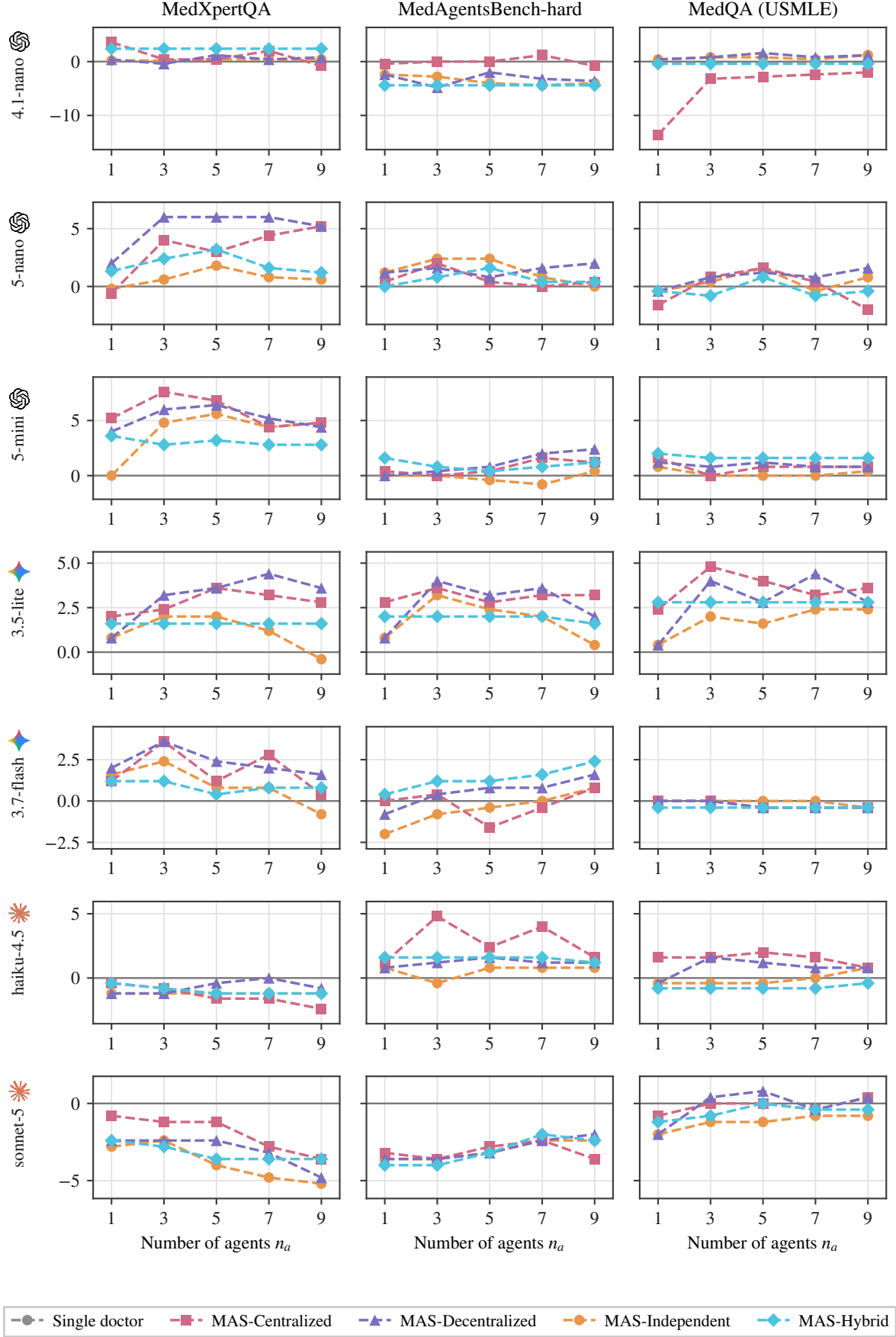


Figure 8: **Number-of-agents scaling.** Gain over the single doctor against panel size $n_a \in \{1, 3, 5, 7, 9\}$, for every architecture, model and benchmark; rows are models, columns benchmarks, and all panels share one vertical scale. Every series stays within a few points of the baseline across the whole range, and where gains occur they are captured by $n_a = 3$.

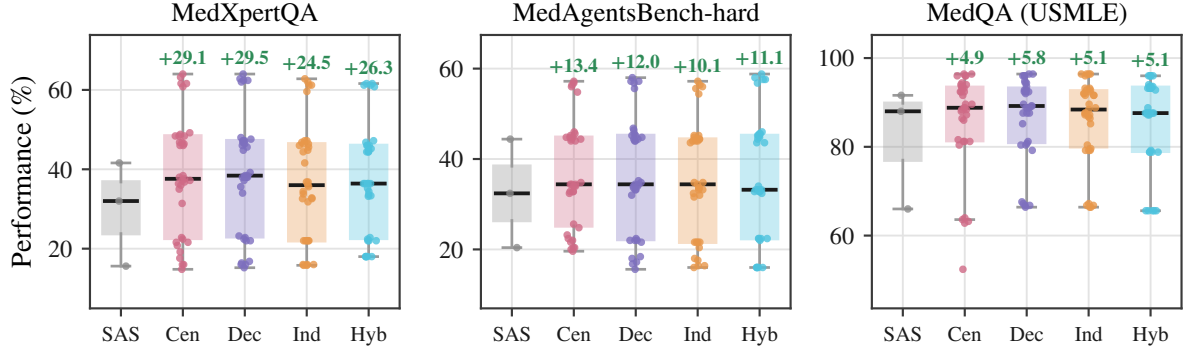


Figure 9: **Single doctor versus panel: performance distributions.** Box plots over all configurations of each architecture, with individual configurations overlaid. Bars are labelled SAS (single doctor), Cen, Dec, Ind and Hyb for Centralized, Decentralized, Independent and Hybrid. Annotations give the change in mean relative to the single doctor, as a percentage of that baseline. The spread within an architecture dwarfs the difference between architectures.

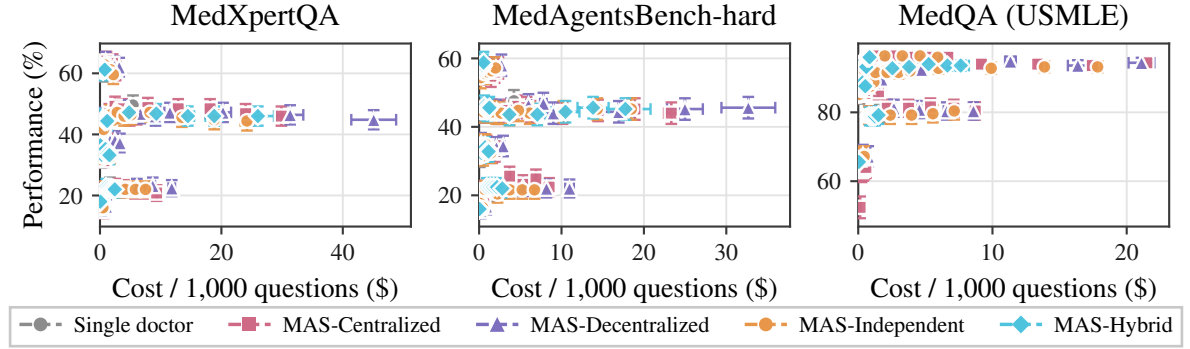


Figure 10: **Cost-performance trade-offs.** Mean performance against nominal cost per 1,000 questions, with SEM error bars on both axes. Each point is one configuration.

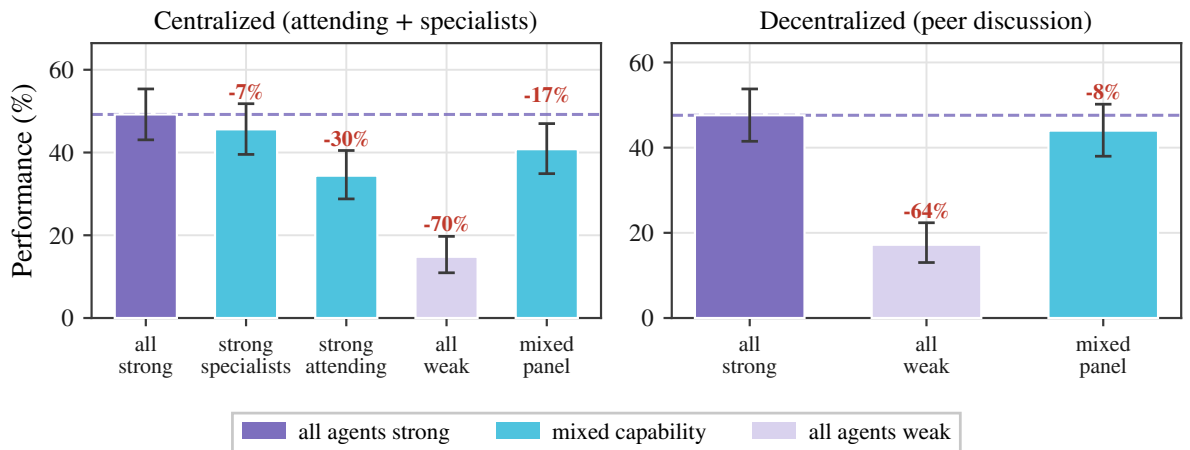


Figure 11: **Agent heterogeneity.** Performance of Centralized and Decentralized panels at $n_a=3$ on MedXpertQA under five capability placements. Strong is gpt-5-mini, weak is gpt-4.1-nano; *strong specialists* pairs a weak attending with strong specialists and *strong attending* reverses that pairing. Annotations give the change relative to the all-strong panel (dashed line). Decentralized coordination has no attending, so the two attending-capability conditions are structurally undefined there and are omitted.